A Method to Estimate Discrete Choice Models That Is Robust to Consumer Search

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We state conditions under which choice data suffice to identify preferences when consumers may not be fully informed about attributes of goods. Our approach can be used to test for full information, forecast how consumers will respond to information, and conduct welfare analysis when consumers are imperfectly informed. In a lab experiment, we successfully forecast the average response to new information when consumers engage in costly search. In data from Expedia, our method identifies which attribute was not immediately visible to consumers in search results and allows us to compute the value of additional information.

I. Introduction

A large literature in marketing and economics documents that consumers are often imperfectly informed about the attributes of relevant products in

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ways that substantially alter their choices (for a review, see Honka, Hortaçsu, and Wildenbeest 2019). This pattern is found for relatively inexpensive items, such as groceries, as well as big-ticket and consequential purchases, such as cars, insurance, and schooling (e.g., Bronnenberg and Vanhonacker 1996; Abaluck and Gruber 2011; Woodward and Hall 2012; Bronnenberg, Kim, and Mela 2016; Allcott, Lockwood, and Taubinsky 2019). Given this, models that assume full information may generate wrong conclusions about welfare and—by construction—cannot be used to assess how consumers would respond to an information intervention.

In this paper, we state what we believe to be plausible sufficient conditions under which choice data alone suffice to recover preferences even if consumers are partially informed. In our baseline model, we assume that if consumers search, they do so in decreasing order of expected utility—a condition we make precise below. We show that if this condition is satisfied, along with a few additional restrictions we describe below, there is a function of choice probabilities that recovers preferences whether consumers are fully or partially informed. Our approach does not require the researcher to fully specify a structural search model beyond the expected utility assumption. Specifically, the expected utility assumption allows for a broad class of search models, including versions of classic sequential search (Weitzman 1979), the "directed cognition" model of Gabaix et al. (2006), or heuristic rules such as satisficing (Simon 1955) and simultaneously searching all goods with expected utility above a threshold.[1]

Recovering preferences under partial information has many applications. First, one can forecast the impact of informing consumers about attributes of goods before conducting such interventions and compute the associated welfare benefits.[2] Second, our approach can inform firms' advertising strategies

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[1] The empirical literature suggests that canonical assumptions in all of these cases are often rejected by the data (in our four examples, Schwartz et al. 2002; Gabaix et al. 2006; Honka and Chintagunta 2017; Jindal and Aribarg 2018), limiting the applicability of fully specified search models.

[2] If an information intervention also reduces search costs, then the welfare gains from better choices can be viewed as a lower bound to the total increase in welfare.

and the interface design of online platforms (e.g., by identifying product attributes that consumers care about but might not currently be aware of).[3] Third, in settings where one would otherwise assume full information, the expected utility assumption provides a generalization that allows for both full and partial information, thus permitting a more reliable normative evaluation of choices.[4] Finally, given preferences recovered by our approach, we show that it is possible to identify other primitives of interest, such as the distribution of search costs under a maintained model of search.

One can think of our approach as a data-driven method of isolating consumers who maximize utility. Consider the example of consumers shopping for laptops on an e-commerce platform. Some product information (e.g., price) is immediately visible on the results page, while other attributes (e.g., shipping speed) are relegated to the product page and require some time and effort to be uncovered; we refer to these attributes as *hidden*. Consumers may fail to maximize utility if they do not pay the cost to learn shipping speed for all products. Our expected utility assumption states that if you bother to check the shipping speed for a given laptop, you will also do so for any other laptop in the choice set that appears more promising to you based on the information immediately visible on the results page. This assumption implies that consumers who search the laptop with the best shipping speed always choose the product that maximizes utility among all options (which is not necessarily that with the best shipping speed). To see this, note that if some other laptop has higher utility than that with the best shipping speed, it must have higher expected utility and thus our assumption implies that it is searched and then chosen by the consumer. Further, only consumers who search the laptop with the best shipping speed are sensitive to shipping speed for that laptop. Therefore, by looking at how the choice probability for the laptop with the best shipping speed changes with that attribute, we are able to isolate consumers who behave as if they were fully informed; standard arguments for the full-information case then recover their preferences.

Based on this argument, we show that a specific ratio of second derivatives of choice probabilities identifies preferences given partial information. In contrast, standard methods, based on ratios of first derivatives, tend to lead to attenuated estimates of preferences. In the above example, those approaches may erroneously conclude that consumers do not care much about shipping speed when in fact they are simply not aware that products differ along that dimension. Our second-derivative ratio recovers preferences even with partial information, and it also recovers preferences in the

[3] This is related to the literature on advertising content (e.g., Anderson, Ciliberto, and Liaukonyte 2013).

[4] For example, when quantifying the gains from improving school quality (as measured by test scores), willingness to pay from actual choices may understate willingness to pay for parents who observe quality (Hastings and Weinstein 2008).

full-information case. Thus, it provides robust estimation whether consumers are fully or partially informed. Further, we extend our model in several empirically relevant directions, including cases where the expected utility assumption is not satisfied. Specifically, we consider extensions where (i) search costs vary with observables (e.g., rank on a web page), (ii) search reveals information that is unobservable to the researcher, (iii) either x or z is endogenous and valid instruments are available, and (iv) an outside option is available with utility known before search.

Our identification proof lends itself naturally to estimation and testing. If one can nonparametrically estimate choice probabilities as a function of product attributes, then our results can be used to directly recover preferences (Compiani 2022). We also suggest an alternative, more parsimonious approach to estimate second derivatives that works well in simulations for larger numbers of goods. Further, one can use our result to test for full information by checking whether our "search-robust" estimates of preferences are equal to the conventional estimates based on first derivatives. This implies that one does not need to take an a priori stance on whether an attribute is uncovered only after searching a good. That hypothesis can be tested provided the data contain at least one attribute that can be assumed to be known by consumers before search.

We conduct two data analyses to validate the usefulness of our approach, assuming homogeneous preferences for product attributes. First, in a lab experiment involving e-book choice, we show that we can recover full-information preferences over hidden discounts using only data on observed choices. Second, using data from hotel choice on Expedia, we show that our method correctly identifies as hidden the one attribute not immediately visible in search results (hotel location).

While our approach gains generality in relaxing the assumption that consumers are perfectly informed, it is potentially limited in the forms of unobserved heterogeneity it can accommodate. In particular, our main argument maintains that consumers have homogeneous preferences for product attributes and thus assumes that individuals who search the good with the highest value of the hidden attribute do not systematically differ in their unobserved preferences relative to the rest of the population. We show that identification does not rely completely on the assumption of unobservably homogeneous preferences, as our result can be extended to random coefficients models, provided we impose the (potentially strong) restriction of monotonicity—that is, that everyone either likes or does not like the hidden attribute. Even with this identifying assumption, estimation of the random coefficients distributions requires parametric restrictions or estimation of derivatives of order higher than two, which may be infeasible in many applications given data limitations. We note that allowing for observed heterogeneity (e.g., including interactions between product attributes and consumer demographics) is instead straightforward.

Estimation of our baseline model involves taking second-order derivatives of flexibly estimated functions as well as evaluating those functions at specific points in their domains, which could lead to slower-than-parametric convergence rates. Thus, we see it as most useful in settings with large datasets. Example applications might include consumer product choice (evaluating whether shoppers are aware of per-unit prices when choosing among options of different sizes), school choice (evaluating whether parents are informed about average test scores or dropout rates for alternative schools), health plan choice (evaluating whether consumers are informed about plan network breadth), financial choice (evaluating whether consumers are informed about attributes of loans—such as the term length—that cannot be easily translated into dollar values), or housing choice (evaluating whether consumers are informed about measurable attributes, such as the noise level in a particular location). In all these cases, our model could also be used to test the impact of information interventions aimed at raising awareness of particular attributes. Of course, the limitations of our assumptions also apply; for example, we might think it plausible that all students prefer schools with lower dropout rates, but some students might also prefer schools with lower test scores due to tracking considerations. If so, our main approach that assumes homogeneous preferences could give biased results for that attribute.

Our result relates to several existing literatures. A theoretical and empirical literature models consumers as choosing from a possibly strict subset of the options available, their "consideration set."[5] Our model differs from much of the consideration set literature in that we consider the complementary problem of imperfect information at the level of attributes rather than goods. Specifically, in our setting, consumers are assumed to costlessly know some information for all goods, implying that, in general, consumer behavior will be affected by the visible attributes of all products, even those that are not searched.[6] A growing literature, including Mehta, Rajiv, and Srinivasan (2003), Kim, Albuquerque, and Bronnenberg (2010,

[5] Roberts and Lattin (1991), Goeree (2008), Conlon and Mortimer (2013), and Gaynor, Propper, and Seiler (2016)—among others—estimate preferences when consumers may only consider some alternatives. Manzini and Mariotti (2014) establish that one can recover consideration probabilities as well as preferences if the data contain choices from every possible subset of the feasible set of goods. Abaluck and Adams (2017) show that identification can be achieved even without this type of variation under certain models of consideration set formation. Cattaneo et al. (2020) and Barseghyan et al. (2021) study partial identification of a general model with heterogeneous consideration sets. Agarwal and Somaini (2022) propose a model in which consumers choose from choice sets that are unobserved to the researcher due to information frictions or supply-side rationing and show how instrumental variables enable identification of preferences, as well as the rule determining which products are included in the choice set.

[6] The recent theoretical literature on this question includes Branco, Sun, and Villas-Boas (2012), Ke, Shen, and Villas-Boas (2016), and Gabaix (2019).

2017), Honka and Chintagunta (2017), Ursu (2018), and Gardete and Hunter (2020), models consumers as searching products to uncover some of their attributes. We show that preferences can be recovered under our assumptions without committing to one specific structural search model.[7] This is similar in spirit to Compiani et al. (2024), who study optimal rankings of products on e-commerce platforms; their model maintains the assumption of sequential search but is flexible as to what components of utility consumers learn via search. Another related literature studies whether consumers make informed choices by comparing the choices of regular consumers with that of a more informed subgroup. Bronnenberg et al. (2015) ask whether pharmacists make similar prescription drug choices to consumers, Handel and Kolstad (2015) ask whether better-informed consumers make different health insurance choices, and Johnson and Rehavi (2016) study whether physicians treat their patients differently when those patients are other physicians. Rather than identifying informed consumers, our paper develops a data-driven way of identifying consumers who maximize utility (despite not necessarily searching all goods) and whose choices can thus be used to recover preferences.

The rest of the paper is organized as follows. Section II lays out our formal framework and proves our key identification results, section III considers several empirically important extensions such as endogenous attributes, section IV discusses the model assumptions and their testability and the (counterfactual) questions that can be addressed using our approach, section V provides details of estimation, section VI offers a guide to practitioners, sections VII and VIII report results from our experiment and Expedia application, respectively, and section IX concludes.

II. Model and Identification

A. *Setup*

There are $J \geq 2$ goods indexed by $j = 1, \ldots, J$ with attributes x_j observed by consumers and the econometrician and an attribute z_j observed by the

[7] In this sense, our paper is also related to the literature on "attribute nonattendance." There is one special case where the problem of imperfect information about attributes has often been addressed in the existing literature. This is the case in which all attributes can be expressed in dollar terms. For example, consumers should not care whether a health insurance plan saves them $100 in premiums or out-of-pocket costs (see Abaluck and Gruber 2011) or whether a light bulb saves them money in up-front costs or shelf life (as in Allcott and Taubinsky 2015). If one dollar-equivalent attribute is assumed to be visible to consumers, it can provide a benchmark for how consumers should respond to a hidden dollar-equivalent attribute. However, in many cases attributes cannot easily be translated into dollars without first estimating consumer preferences. In these cases, our results still allow one to recover preferences given imperfectly informed consumers.

econometrician but not necessarily by consumers.[8] To focus on the intuition underlying our key results, we make five simplifying assumptions in our exposition here. First, we let x_j be scalar for all j; our results immediately extend to the case of vector-valued x_j's at the cost of some extra notation. Second, we focus on the case where z_j is also a scalar; we consider the case with multivariate z_j in section A.3 of the appendix. Third, we assume that x_j and z_j are continuously distributed (app. sec. A.8 shows that an analogous argument applies to the case with discrete attributes provided that at least one attribute is continuous). Fourth, we assume that the utility that individual i derives from good j is linear in x_j, z_j and an idiosyncratic shock ϵ_{ij} that is observed by consumers before search (in app. sec. A.6, we consider the case where unobservables are seen only after search). Fifth, we focus on the case where both x_j and z_j are exogenous (sec. III.A discusses how to deal with endogeneity). We formalize these assumptions as follows.

Assumption 1. The utility that individual i derives from good j is $U_{ij} = \alpha x_j + \beta z_j + \epsilon_{ij}$, where x_j, z_j are continuous scalars and $\mathbf{x} = (x_1, \dots, x_J)$ and $\mathbf{z} = (z_1, \dots, z_J)$ are independent of $\epsilon_\mathrm{i} = (\epsilon_{i1}, \dots, \epsilon_{iJ})$. The consumer observes x_j, ϵ_{ij} for all j before search but needs to search good j to uncover z_j.

We assume that consumers form expectations on z_j based on x_j: $E(z_j|x_j) = \gamma_0 + \gamma_1 x_j$. Then, the expected utility from good j before engaging in search takes the form $\mathrm{EU}_{ij} = \beta\gamma_0 + (\alpha + \beta\gamma_1)x_j + \epsilon_{ij}$. This nests the case where consumers have (linear) rational expectations as well as the case where they do not update at all about z_j based on observed attributes of goods ($\gamma_1 = 0$). We let $\tilde{z}_j \equiv z_j - (\gamma_0 + \gamma_1 x_j)$ be the surprise utility revealed by searching j relative to consumers' expectations.

Next, we state the assumptions that characterize the class of search models we consider.

Assumption 2.

i) Consumer i searches goods in decreasing order of EU_{ij}.
ii) Conditional on having utility $\bar{u}$ in hand (i.e., having uncovered a maximum utility $\bar{u}$ in the search process so far), consumer i searches j if and only if $g_i(x_j, \epsilon_{ij}, \bar{u}) \geq 0$, where g_i is a decreasing function of $\bar{u}$.[9]

[8] Our model also permits the more general case where attributes are potentially both good and individual specific, but we write x_j and z_j rather than x_{ij} and z_{ij} for notational simplicity.

[9] Assumption 2(ii) can be weakened to allow the function g_i to depend on a good-specific unobservable, such as search costs; however, good-specific search costs may lead to violations of assumption 2(i). In sec. III.B, we extend our model to permit search costs to vary across goods with observable factors.

iii) Consumers choose the good that maximizes utility among searched goods.

iv) Only the value of z_j is unknown to consumers before search, and search fully reveals z_j.

We discuss these conditions at length in section IV.A. To briefly clarify, assumption 2(i) states that consumers search goods in descending order of expected utility before search. Assumption 2(ii) states that consumers decide whether to search a good based on their utility in hand and the information they have available about the good they are considering searching (i.e., x_j, ϵ_{ij}). This rules out, for example, a sequential search protocol whereby one stops searching after discovering a good with large z irrespective of utility in hand.[10] We use a subscript i with the function g to emphasize that the function may depend on any individual (unobserved) heterogeneity in utility or search. For example, in a Weitzman search model, the stopping rule would depend on consumer i's reservation value, which in turn depends on i's search cost. Assumption 2(iii) states that consumers must search a good before choosing it. Assumption 2(iv) states that the econometrician observes all the information that is revealed by search, and that search is fully informative about the hidden attribute.

We pause here to highlight that assumption 2 accommodates several commonly used models of search.

Example 1 (Sequential search). Suppose that consumers search sequentially and consumer i must pay a cost c_i every time she uncovers the z attribute for a good. Further, assume that the consumer believes that $\tilde{z}$ is distributed according to the prior $F_{\tilde{z}}$ independent and identically distributed (i.i.d.) across goods. Then, following Weitzman (1979), the consumer will rank goods according to their reservation value rv'_{ij} defined implicitly by

$$c_i = \int_{\mathrm{rv}'_{ij}}^{\infty} (u - \mathrm{rv}'_{ij})\, dF_{U_{ij}|x_j}(u) = \int_{\mathrm{rv}_i}^{\infty} \beta(t - \mathrm{rv}_i)\, dF_{\tilde{z}}(t), \tag{1}$$

where $\mathrm{rv}_i \equiv (\mathrm{rv}'_{ij} - \beta\gamma_0 - (\alpha + \beta\gamma_1)x_j - \epsilon_{ij})/\beta$ and the last step follows from a change of variable. We can interpret rv_i as the reservation value in units of $\tilde{z}$. To see this, note that consumer i searches goods in descending order of $\beta\gamma_0 + (\alpha + \beta\gamma_1)x_j + \beta\mathrm{rv}_i + \epsilon_{ij}$ (or, equivalently, of $\mathrm{EU}_{ij} = \beta\gamma_0 + (\alpha + \beta\gamma_1)x_j + \epsilon_{ij}$), and for each good j' she chooses to uncover $z_{j'}$

[10] Assumption 2(ii) does accommodate simultaneous search models in which consumers decide which goods to uncover based on expected utilities and then proceed to jointly search them. In this case, consumers do not sequentially uncover utilities and the function g_i does not vary with its third argument.

if and only if the maximum utility secured so far is lower than $\beta\gamma_0 + (\alpha + \beta\gamma_1)x_{j'} + \beta \mathrm{rv}_i + \epsilon_{ij'}$. Once she stops searching, she maximizes utility among the searched goods. Thus, assumption 2 is satisfied with $g_i(x_j, \epsilon_{ij}, \bar{u}) = \beta\gamma_0 + (\alpha + \beta\gamma_1)x_j + \beta \mathrm{rv}_i + \epsilon_{ij} - \bar{u}$.

EXAMPLE 2 (Directed cognition model). As in the model of Gabaix et al. (2006), suppose that consumers rank goods in terms of expected utility and myopically check whether searching the next good is worth the cost. The directed cognition model has the same search order as the Weitzman model but a different g_i function; consumers myopically check whether $g_i(x_j, \epsilon_{ij}, \bar{u}) = E_{\tilde{z}} \max(U_{ij} - \bar{u}, 0) - c_i \geq 0$ to decide whether to search good j.

EXAMPLE 3 (Satisficing). Suppose that consumer i searches in order of expected utility and stops whenever utility in hand is above a threshold τ_i. Then, assumption 2 is satisfied with $g_i(x_j, \epsilon_{ij}, \bar{u}) = \tau_i - \bar{u}$.

EXAMPLE 4 (Full information). The full-information model is subsumed within the previous example by letting $\tau_i = \infty$ for all i.

EXAMPLE 5 (Simultaneous search). Suppose that consumer i simultaneously searches all goods that have expected utility above a threshold $\tilde{\tau}_i$. Then, assumption 2 is satisfied with $g_i(x_j, \epsilon_{ij}, \bar{u}) = \beta\gamma_0 + (\alpha + \beta\gamma_1)x_j + \epsilon_{ij} - \tilde{\tau}_i$.

Our results will not require the researcher to take a stand on the specific model of search that consumers follow provided that our assumptions are met. Therefore, as illustrated by the examples above, the approach will be agnostic as to whether consumers search sequentially or simultaneously, are forward looking or myopic, and have biased or unbiased beliefs, among other things. In contrast, fully specifying a structural model requires one to take a stance on each of these dimensions.

B. *Identification*

We assume throughout without loss that $\beta > 0$—that is, we treat z_j as an attribute that consumers value in good j.[11] Further, we let product 1 be the good with the highest value of $\tilde{z}$. In other words, good 1 delivers the largest positive surprise relative to consumers' expectations. Note that $\tilde{z}_j$

[11] This is without loss, since assumption 2 implies that an increase in U_{ij} can only induce consumer i to switch from not choosing j to choosing j but never vice versa (see n. 34 for a formal argument). Thus, by the chain rule, the sign of β is identified by the sign of $\partial s_j/\partial z_j$, where s_j is the choice probability function for good j from the data.

collapses to z_j when $\gamma_1 = 0$—that is, when consumers do not form any expectations on z (so in the case of myopic expectations, good 1 simply has the highest value of z). Here we consider the case where the parameter γ_1 governing the way in which consumers form expectations is known to the researcher. For instance, if consumers have rational expectations, γ_1 can simply be estimated by regressing z_j on x_j. Knowledge of γ_1 implies knowledge of $\tilde{z}_j$ for all j, which means that the researcher knows the identity of good 1 in any given choice set. Section III.E considers the case where γ_1 is unknown to the researcher.

We are now ready to state and prove a key lemma.

LEMMA 1. Let assumptions 1 and 2 hold. If consumer i searches good 1 (i.e., the good with the highest value of $\tilde{z}$), then i chooses the utility-maximizing good.

Proof. If good 1 is searched but utility is not maximized, then for some unsearched j, $U_{ij} > U_{i1}$. Since $\tilde{z}_1 \geq \tilde{z}_j$, it must be that $\mathrm{EU}_{ij} > \mathrm{EU}_{i1}$. But by assumption 2(i), this implies that good j is searched, which is a contradiction. QED

Note that lemma 1 does not imply that good 1 always maximizes utility if it is searched. Rather, it implies that if good 1 is searched, the utility-maximizing good will also be searched (regardless of whether it is good 1) and thus the consumer will choose that good. The lemma also does not mean that consumers searching good 1 are fully informed (in a search model, they typically will not be) but simply that those consumers act *as if* they were fully informed.

Lemma 1 will have far-reaching implications. In particular, it implies that z_1 impacts only the choice probability for good 1, denoted s_1, for individuals who behave as if they were fully informed. Therefore, looking at $\partial s_1/\partial z_1$ will isolate individuals who maximize utility and allow us to recover preferences using standard full-information arguments. To formalize this, we write the choice probability for good 1 as[12]

$$\begin{aligned} s_1 = {} & P\,(\text{good 1 maximizes utility among all goods}) \\ & - P\,(i \text{ fails to choose 1 when it maximizes utility}) \\ & + P\,(i \text{ chooses 1 even though it does not maximize utility}). \end{aligned}$$

[12] Note that this probability is computed by integrating over any individual-specific unobserved heterogeneity in utility or search, while fixing the choice set attributes $\mathbf{x} \equiv [x_1, \ldots, x_J]$ and $\mathbf{z} \equiv [z_1, \ldots, z_J]$. Therefore, s_1 is a function of $(\mathbf{x}, \mathbf{z})$, but we will often omit the dependence from the notation. Throughout the paper, the sources of individual-level unobserved heterogeneity will vary with the specific models we consider, so the symbol P denotes integrals over different distributions depending on the context. For instance, in the Weitzman model of example 1, the probability is taken over the joint distribution of the preference shocks ϵ_i and the search costs c_i.

In words, the choice probability of good 1 equals the full-information probability adjusted for two types of mistakes that imperfectly informed consumers may make. Lemma 1 implies that the second type of mistake (choosing 1 even though it does not maximize utility) never happens, so that we can write

$$s_1 = P(U_{i1} \geq U_{ik} \;\; \forall\, k) - P\Big(\{U_{i1} \geq U_{ik} \;\; \forall\, k\} \cap \{\text{for some } j \neq 1, \text{EU}_{ij} \geq \text{EU}_{i1} \text{ and } g_i(x_1, \epsilon_{i1}, U_{ij}) \leq 0\}\Big), \tag{2}$$

where we use that, in our model, failing to search good 1 requires that there exists some other good j with $\text{EU}_{ij} \geq \text{EU}_{i1}$ and utility high enough that $g_i(x_1, \epsilon_{i1}, U_{ij}) \leq 0$.

We now use equation (2) to show our key result—that the parameters of the utility distribution are identified from the second derivatives of function s_1. Since the distribution of ϵ is unrestricted, we impose the following normalizations without loss: $\alpha = 1$ (scale) and $\epsilon_{i\tilde{j}} = 0$ for some $\tilde{j}$ and all i (location).

LEMMA 2. Let assumptions 1 and 2 hold. Further, assume that s_1 is twice differentiable and that $(\partial^2 s_1/(\partial z_1 \partial x_j))(\mathbf{x}^*, \mathbf{z}^*) \neq 0$ for some $(\mathbf{x}^*, \mathbf{z}^*)$ and some $j \neq 1$. Then,

$$\frac{\partial^2 s_1}{\partial z_1 \partial z_j}(\mathbf{x}^*, \mathbf{z}^*) \Big/ \frac{\partial^2 s_1}{\partial z_1 \partial x_j}(\mathbf{x}^*, \mathbf{z}^*) = \frac{\beta}{\alpha} = \beta, \tag{3}$$

so that β is identified. In addition, the distribution of ϵ is nonparametrically identified if $\text{supp}(\epsilon_k - \epsilon_{\tilde{j}})_{k \neq \tilde{j}} \subset \{(\alpha + \beta\gamma_1)(x_{\tilde{j}} - x_k)_{k \neq \tilde{j}} : \gamma_1(x_{\tilde{j}} - x_k)_{k \neq \tilde{j}} = (z_{\tilde{j}} - z_k)_{k \neq \tilde{j}}$ for some $(\mathbf{x}, \mathbf{z})$ in its support$\}$ and the support of $(\tilde{z}_1, \ldots, \tilde{z}_J)$ contains a point such that $\tilde{z}_j = \tilde{z}_k$ for all j, k.

Proof. To facilitate intuition, we focus on the case with $J = 2$ goods here. Appendix section A.1 contains the proof for the general case with $J \geq 2$ goods. First, we prove equation (3), often suppressing the subscript i in what follows. Let $\tilde{u}_j \equiv \alpha x_j + \beta z_j$ be the mean utility of good j, so that $U_{ij} = \tilde{u}_j + \epsilon_{ij}$, and let $(\mathbf{x}, \mathbf{z}) = (\mathbf{x}^*, \mathbf{z}^*)$. Our goal is to show that both z_2 and x_2 impact $\partial s_1/\partial z_1$ only via $\tilde{u}_2$. This in turn implies that $\partial^2 s_1/(\partial z_1 \partial z_2) = (\partial^2 s_1/(\partial z_1 \partial \tilde{u}_2))(\partial \tilde{u}_2/\partial z_2)$ and $\partial^2 s_1/(\partial z_1 \partial x_2) = (\partial^2 s_1/(\partial z_1 \partial \tilde{u}_2))(\partial \tilde{u}_2/\partial x_2)$, implying the result in equation (3).

To this end, note that with two goods, the second term in (2) takes the form

$$P(\{U_1 > U_2\} \cap \{\text{EU}_2 > \text{EU}_1\} \cap \{g(x_1, \epsilon_1, U_2) \leq 0\}) = P(\{U_1 > U_2\} \cap \{g(x_1, \epsilon_1, U_2) \leq 0\}) - P(\{\text{EU}_1 > \text{EU}_2\} \cap \{g(x_1, \epsilon_1, U_2) \leq 0\}), \tag{4}$$

where the second line follows since $\mathrm{EU}_1 > \mathrm{EU}_2$ implies $U_1 > U_2$ and thus $P(\{\mathrm{EU}_1 > \mathrm{EU}_2\} \cap \{g(x_1, \epsilon_1, U_2) \le 0\}) = P(\{U_1 > U_2\} \cap \{\mathrm{EU}_1 > \mathrm{EU}_2\} \cap \{g(x_1, \epsilon_1, U_2) \le 0\})$. The second term on the last line of display (4) is not a function of z_1. The first term is a function of x_2 and z_2 only via $\tilde{u}_2$. This implies that both z_2 and x_2 impact $\partial s_1/\partial z_1 = (\partial P(U_1 > U_2))/\partial z_1 - (\partial P(\{U_1 > U_2\} \cap \{\mathrm{EU}_2 > \mathrm{EU}_1\} \cap \{g(x_1, \epsilon_1, U_2) \le 0\})/\partial z_1)$ only via $\tilde{u}_2$. Together with the normalization $\alpha = 1$, this proves equation (3).

We now show that the distribution of ϵ can be identified. Consider choice sets where $\tilde{z}_j$ is the same for all j. In this case, the good with the highest expected utility also has the highest realized utility. Thus, consumers who search in descending order of expected utility always choose the utility-maximizing good, and we can write

$$s_{\tilde{j}} = P(\epsilon_k - \epsilon_{\tilde{j}} \le (\alpha + \beta\gamma_1)(x_{\tilde{j}} - x_k) \ \ \forall\ k). \tag{5}$$

This is a standard full-information demand model, and we can follow the argument in section IV.B of Berry and Haile (2014) to identify the joint distribution of ϵ. Specifically, letting $\mathbf{x}$ vary—and relying on the fact that the coefficient $\alpha + \beta\gamma_1$ is identified given knowledge of γ_1—one can use (5) to trace out the distribution of $(\epsilon_k - \epsilon_{\tilde{j}})_{k \neq \tilde{j}}$ over the set in the statement of the theorem. This, paired with the location normalization $\epsilon_{i\tilde{j}} = 0$, gives identification of the distribution of ϵ. QED

This result shows that the preferences parameter β is identified from our ratio of second derivatives and that the entire distribution of ϵ can be traced out nonparametrically under additional support restrictions. Specifically, this second argument relies on varying $\mathbf{x}$ and $\mathbf{z}$ in such a way that $\tilde{z}_j = \tilde{z}_k$ for all j, k, which is equivalent to $\gamma_1(x_j - x_k) = z_j - z_k$. Thus, we require the support of ϵ to be contained in the set of $(\alpha + \beta\gamma_1)\mathbf{x}$ values that satisfy this condition (for some value of $\mathbf{z}$). Note that in the case where $\gamma_1 = 0$, the assumption reduces to requiring that the support of ϵ be contained in the support of $\mathbf{x}$, which is in line with restrictions used in the literature (Berry and Haile 2014). In practice, nonparametrically estimating the distribution of ϵ may be challenging with limited data. Indeed, in our empirical applications we leverage standard parametric assumptions. As a separate remark, because the argument that identifies the distribution of ϵ conditions on the slice of the data where $\tilde{z}_j$ is the same for all goods, it is likely to lead to slower than parametric convergence rates (Khan and Tamer 2010). If—as in our applications—the distribution of ϵ is assumed to be known and thus α cannot be normalized without loss, a similar caveat applies to estimating α.

Finally, we show that in many models of interest, identifying preferences based on the ratio of first derivatives leads to understating consumers' taste for z. For this result, we make two mild additional assumptions: (i) that the function $g_i(x_j, \epsilon_{ij}, \bar{u})$ is weakly increasing in x_j and (ii) that

$\alpha + \beta\gamma_1 \geq 0$. Condition i is satisfied in all the search models considered above (examples 1–5) when the coefficient on x in utility is positive and corresponds to the mild requirement that consumers are (weakly) more prone to searching a good the higher the value of x for that good. Condition ii is also intuitive and requires that the effect of x_j through expectations about z_j (i.e., $\beta\gamma_1$) not flip the sign of the overall effect. For example, if x_j captures how inexpensive the product is and z_j is its quality, then it must be that consumers still value a decrease in price even after accounting for the fact that they expect lower quality.

Lemma 3. Let assumptions 1 and 2 hold. Further, assume that $g_i(x_j, \epsilon_{ij}, \bar{u})$ is weakly increasing in x_j and that $\alpha + \beta\gamma_1 \geq 0$. Then, for all j, k,

$$\left| \frac{\partial s_j}{\partial z_k} \Big/ \frac{\partial s_j}{\partial x_k} \right| \leq |\beta|. \tag{6}$$

Proof. See appendix section A.2.

The results in lemmas 2 and 3 suggest a natural approach to test for full information. Under the null hypothesis of full information, $s_j = P(U_{ij} \geq U_{ik} \;\; \forall\, k)$ and therefore, by the chain rule,

$$\frac{\partial s_j}{\partial z_k} \Big/ \frac{\partial s_j}{\partial x_k} = \frac{\partial^2 s_1}{\partial z_1 \partial z_{j'}} \Big/ \frac{\partial^2 s_1}{\partial z_1 \partial x_{j'}} = \beta \tag{7}$$

for all j, k and all $j' \neq 1$. On the contrary, when consumers are unaware of z for some goods, then the ratios of first derivatives need not be equal to the ratios of the second derivatives. In particular, lemma 3 showed that the ratios of first derivatives will tend to be smaller in magnitude within the class of search models we consider. This motivates the testing approach in the following lemma.

Lemma 4. Under assumption 1, if consumers are fully informed, then $|(\partial s_j/\partial z_k)/(\partial s_j/\partial x_k)| = |(\partial^2 s_1/\partial z_1 \partial z_{j'})/(\partial^2 s_1/\partial z_1 \partial x_{j'})|$ for all j, k and all $j' \neq 1$. Under the assumptions of lemma 3, $|(\partial s_j/\partial z_k)/(\partial s_j/\partial x_k)| \leq |(\partial^2 s_1/\partial z_1 \partial z_{j'})/(\partial^2 s_1/\partial z_1 \partial x_{j'})|$ for all j, k and all $j' \neq 1$. Thus, it is possible to test the hypothesis of full information by checking whether the ratios of first derivatives are attenuated relative to the ratios of second derivatives.

A natural question is whether the testable implications in lemma 4 are still valid when assumption 1 is violated and in particular when the coefficients on product attributes are heterogeneous across consumers. One might worry that when consumers have heterogeneous preferences, the ratios of first derivatives might be attenuated relative to the ratios of second derivatives even under the null hypothesis of full information. In section III of the supplementary appendix (available online), we provide verifiable sufficient conditions that rule this out when the ϵ shocks are Gumbel distributed and the coefficients can be signed (e.g., β is distributed on the positive reals); the latter is needed to ensure that the identity of good 1 is consistent across consumers. Under these conditions, the testing approach based on second derivatives is valid even in the presence of heterogeneity in α or β. Further, we show that this is not generally the case when $\beta = 0$ for a positive mass of consumers. Intuitively, if some consumers do not care about z at all, this will tend to lead to attenuation of the first-derivative ratios even under full information. However, we show that using a slightly different approach allows us to distinguish between this scenario and the case where consumers search (and have homogeneous β). More specifically, we use the fact that ratios of derivatives of the same order (i.e., $(\partial^2 s_1/\partial z_1 \partial z_2)/(\partial^2 s_1/\partial z_1 \partial x_2)$ and $(\partial^2 s_2/\partial z_2 \partial z_1)/(\partial^2 s_2/\partial z_2 \partial x_1)$) are equal under the assumption of full information, even when $\beta = 0$ for some consumers. In contrast, these ratios will in general not be the same when consumers engage in search. This discrepancy allows us to empirically distinguish the two cases.

So far we have discussed how to adjust our testing approach to accommodate unobserved heterogeneity in β on the positive or negative reals. Identifying the distribution of the random coefficients in our setting is more challenging and in general will require taking higher-order derivatives or maintaining parametric assumptions on the distribution of the random coefficients. We provide some results on this in section I.1 of the supplementary appendix.

III. Extensions

We now consider a few extensions to our baseline model that are relevant for empirical work.

A. *Endogenous Attributes*

So far, we have assumed that the observed product attributes are independent of all unobservables. This can be restrictive, especially in settings in which product attributes—notably, price—are chosen by firms who might know more about preferences or product attributes than is captured by the observed data. As highlighted by a large literature (e.g., Berry, Levinsohn, and Pakes 1995), this typically leads to correlation between the attributes chosen by firms and product-level unobservables.

Here we consider an extension of our model that allows for endogenous product attributes. In each market, we assume that there is a continuum of consumers indexed by i and specify the utility they get from good j as

$$U_{ij} = \alpha x_j + \beta z_j + \lambda p_j + \xi_j + \epsilon_{ij}, \tag{8}$$

where p_j denotes the endogenous characteristic and ξ_j is a product-specific characteristic that is known by consumers before search but is not observed by the researcher.

We consider both the case where p_j is visible to consumers without search and that in which consumers need to search good j to uncover p_j (in which case, with a slight abuse of notation, p_j coincides with z_j). If p_j is price, the first scenario corresponds to settings such as e-commerce where price is typically visible on the results page and does not require any further clicking by the user. Alternatively, the second scenario covers cases in which price is itself the object of consumer search (there is a large literature on this, particularly in relation to the often observed price dispersion for relatively homogeneous goods; see, e.g., Stahl 1989; Hortaçsu and Syverson 2004; Hong and Shum 2006). We show identification of preferences for each of these two cases. To this end, we introduce two mutually exclusive variants of assumptions 2(i) and 2(ii).

Assumption 3.

i) The attribute p_j is visible to consumers before search, and consumers form expectations on z_j using $E(z_j|x_j, p_j) = \gamma_0 + \gamma_1 x_j + \gamma_{1,p} p_j$. Further, consumers search in descending order of expected utility $\mathrm{EU}_{ij} = \delta_j + \epsilon_{ij}$, where $\delta_j \equiv \beta\gamma_0 + (\alpha + \beta\gamma_1)x_j + (\lambda + \beta\gamma_{1,p})p_j + \xi_j$. Conditional on having utility $\bar{u}$ in hand, consumer i searches j if and only if $g_i(\delta_j, \epsilon_{ij}, \bar{u}) \geq 0$, where g_i is increasing in δ_j and decreasing in $\bar{u}$.
ii) The attribute p_j is uncovered by consumers only upon searching good j. In this case, z_j coincides with p_j and consumers form expectations on p_j using $E(p_j|x_j) = \gamma_0 + \gamma_1 x_j$. Further, consumers search in descending order of expected utility $\mathrm{EU}_{ij} = \underline{\delta}_j + \epsilon_{ij}$, where $\underline{\delta}_j \equiv \beta\gamma_0 + (\alpha + \beta\gamma_1)x_j + \xi_j$. Conditional on having utility $\bar{u}$ in hand, consumer i searches j if and only if $g_i(\underline{\delta}_j, \epsilon_{ij}, \bar{u}) \geq 0$, where g_i is increasing in $\underline{\delta}_j$ and decreasing in $\bar{u}$.

Like assumptions 2(i) and 2(ii), assumption 3 states that consumers search in descending order of expected utility and decide whether to search good j based on the utility in hand and the information they have on good j before search. In appendix section A.4, we show that our model falls within the class of demand systems studied by Berry and Haile (2014). Thus, we can invoke their results to establish nonparametric identification of the structural choice probability functions—that is, the functions that map all market-level attributes, including the unobservables ξ, into the probability of purchase. Of course, this requires valid instruments for the endogenous variables, but the requirements are no more demanding than in the case of full information. In particular, standard arguments can be applied to motivate the usual instruments for price, such as cost shifters, exogenous attributes of competing products, and prices of the same products in other markets. One caveat is that to ensure that the exclusion restriction is satisfied, it is necessary to assume that consumers do not know these instruments or, if they do, that they do not use them to form expectations on z. For instance, one might assume that consumers are not aware of production cost shifters when making their choices. Once the structural choice probability functions are identified, one may apply our results in section II.B to identify the preference parameters in (8). Note that, as in the case with full information, recovering the structural probability functions when some of the product attributes are endogenous requires variation across many markets. This is more demanding of the data relative to the case without endogeneity where only a few markets suffice in principle (see sec. IV.B for more discussion of this point).

B. *Allowing for Variables Affecting Search but Not Utility*

One important case in which the assumption that consumers search in descending order of expected utility (assumption 2[i]) is likely to fail is when some factors impact search costs but not utility. An example might be search position for online purchases. Arguably, search position impacts the order in which people search but often has no direct impact on utility conditional on search (Ursu 2018). In this case, consumers might first search items with higher search positions even if they do not have higher expected utility. For example, if we randomly assign search order, this is likely to impact choices even though we are not changing the utility of each item conditional on search. Another example is advertising, which entices consumers to search advertised goods but may not affect their utility.

Our model can be extended to deal with cases where the factors impacting search but not utility are observable and the sign of their impact on search probabilities is known (such as position in a list of e-commerce results). Denoting this variable by r_j, suppose that r_j is observed by the researcher and that higher values of r_j make a good weakly more likely to be searched. Now, rather than assuming that goods are searched based on EU_{ij} alone, we assume that goods are searched based on $m(\mathrm{EU}_{ij}, r_j)$, where m is strictly increasing in both EU_{ij} and r_j. In appendix section A.5, we show that our identification argument from lemma 2 continues to hold under additional assumptions, including an independence of irrelevant alternatives (IIA) restriction. In spite of these additional, stronger assumptions, we show that our results are not sensitive to implementing this alternative model in our empirical application in section VIII. Last, the appendix shows that identifying the distribution of ϵ requires the support of $(r_1, \ldots, r_J)$ to contain a point where r_j is the same across all goods, which rules out the case where r is search position. However, this support restriction is not needed to identify the marginal rate of substitution between x and z or test for full information.

C. *Unobservables Revealed by Search*

Our baseline model focused on the case where the attribute(s) z revealed by searching a good are entirely observed by the researcher. However, it is easy to imagine settings in which the data do not capture all of the information that consumers acquire through search. Indeed, the existing literature often models search as the process whereby the idiosyncratic preference shocks—ϵ_{ij} in our notation—are revealed (e.g., Kim, Albuquerque, and Bronnenberg 2010; Ursu 2018; Moraga-González, Sándor, and Wildenbeest 2021). To accommodate this, we consider a modification of our model where the shock ϵ_{ij} becomes known to consumer i only upon searching good j (along with z_j). In other words, consumers know x_j for all j before search and decide whether to acquire ϵ_{ik} and z_k for any given good k through search. As a result, the expected utility in assumption 2(i) now takes the form $\mathrm{EU}_j = \beta\gamma_0 + (\alpha + \beta\gamma_1)x_j$, the rule determining whether a good is searched in assumption 2(ii) is now $g_i(x_j, \bar{u}) \geq 0$, and assumption 2(iv) is dropped.

In this case, the hidden part of utility is not fully observed by the researcher and thus lemma 1 does not hold. However, if consumers tend to value the x attribute (i.e., $\alpha + \beta\gamma_1 \geq 0$), they will search in descending order of it, implying that they will always search the good with the highest x. Using this fact, in appendix section A.6 we show that the ratio of second derivatives $(\partial^2 s_j/\partial z_j \partial z_k)/(\partial^2 s_j/\partial z_j \partial x_k)$ recovers β/α provided that one chooses good k to be the good with the highest value of x, which could coincide with good 1 (here j need not be the good with the highest value of $\tilde{z}$). Recall that in our baseline model, $(\partial^2 s_1/\partial z_1 \partial z_k)/(\partial^2 s_1/\partial z_1 \partial x_k) = \beta/\alpha$ for any choice of $k \neq 1$. Thus, specifically choosing good k to be the good with the highest value of x works both in our baseline model and when unobservables are instead revealed by search rather than being part of expected utility. We estimate this version of the model in our empirical application of section VIII and find that the results are robust. In appendix section A.6, we also show that it is possible to test the hypothesis that ϵ is revealed only via search. Indeed, under this hypothesis, we show that the ratio of first derivatives $(\partial s_1/\partial z_k)/(\partial s_1/\partial x_k)$ also recovers β/α when k is the good with the highest value of x, so that $(\partial s_1/\partial z_k)/(\partial s_1/\partial x_k) = (\partial^2 s_1/\partial z_1 \partial z_k)/(\partial^2 s_1/\partial z_1 \partial x_k)$. Instead, if ϵ is visible before search, in general we have $(\partial s_1/\partial z_k)/(\partial s_1/\partial x_k) \neq (\partial^2 s_1/\partial z_1 \partial z_k)/(\partial^2 s_1/\partial z_1 \partial x_k)$. Thus, testing equality of these two ratios provides a test of the null hypothesis that ϵ is revealed only via search.

D. *Outside Option*

So far, we have implicitly assumed that all products are "inside products," in the sense that they each have x and z attributes observed by the researcher. However, in many settings one may want to model an outside option, which corresponds to choosing none of the products for which attribute data are available. Further, the literature often assumes that consumers know the utility of the outside option for free.[13] First, note that lemma 1 still holds in this case: if consumers search good 1 (now defined as the inside good with the highest value of $\tilde{z}$), they always choose the utility-maximizing good (which could be the outside option). As a consequence, our result can still be applied to estimate β, as we formally show in appendix section A.9.

The presence of the outside option can rationalize the pattern whereby a consumer is faced with a set of options and chooses to not search any of them. In our model, this happens when the utility from the outside option, U_{i0}, is sufficiently high: $g_i(x_j, \epsilon_{ij}, U_{i0}) \leq 0$ for all j. This creates a complication when estimating the distribution of ϵ: consumers might mistakenly choose not to search any of the inside goods, implying that even when we condition on $\tilde{z}_j = \tilde{z}$ for all j, we cannot use full-information arguments to trace out the distribution of ϵ. Following many papers in the search literature, we rule this out by assuming that consumers can search one inside good costlessly (e.g., see Hortaçsu and Syverson [2004] and the review in Ursu, Seiler, and Honka [2023]). This assumption implies that at least one of the inside goods is always searched and can be thought of as a way to restrict attention to the population of consumers who are sufficiently interested in the product category.[14]

Finally, in a model with an outside option, one would typically want to allow for a constant term shifting the utility of the inside products relative to the outside option. This can be captured by including among the x variables a dummy for the inside products.

E. *Expectations Parameter γ_1 Unknown to the Researcher*

Our main results in section II.B assumed that the researcher knows the parameters governing consumers' expectations on z. We now consider the case in which the researcher is not willing to assume a value of γ_1. While studying identification of the way consumers form expectations is beyond the scope of this paper, we show that we can still gain some traction under

[13] Alternatively, if one assumes that consumers need to search to uncover the utility of the outside option, then the latter is no different from any other good and thus our arguments from sec. II.B immediately apply as long as there are at least two "inside" goods—i.e., goods for which variation in x and z is available in the data.

[14] When this assumption is violated, the direction of the resulting misspecification bias will in general depend on the specific search protocol that consumers use. We discuss this further, including the special case where the sign of the bias can be signed, in app. sec. A.9.

additional assumptions. Specifically, suppose that the sign of γ_1 is known (e.g., higher-priced goods are of higher quality). Without loss, we assume that $\gamma_1 > 0$. In addition, suppose that there exist choice sets in which a good has both the highest value of z and the lowest value of x. Even when γ_1 is unknown, this good is known to maximize the residual $\tilde{z}_j = z_j - \gamma_1 x_j$, and we label it by 1. With good 1 defined appropriately, the argument from lemma 2 shows that second derivatives with respect to z_1, z_j, x_j for $j \neq 1$ identify β/α, which simply equals β given the normalization $\alpha = 1$. Note that, unlike in the case with γ_1 known, now we cannot in general recover the distribution of ϵ since this requires varying $\mathbf{x}$ while keeping $\tilde{z}_j$—which is a function of the unknown γ_1—fixed at a common value for all j. Therefore, this result is not sufficient to simulate choices with full information. However, comparing the ratio of second derivatives with the ratio of first derivatives does allow us to conduct tests for full information as discussed above.

IV. Discussion

A. *Discussion of Search Model Assumptions*

As discussed above, there are several microfoundations for the first assumption. For example, in the Weitzman (1979) search model, consumers search goods in order of reservation utility, which is a function of the product attributes that are visible before search, the distribution of the hidden attribute $\tilde{z}_j$, and search costs. If $\tilde{z}_j$ is i.i.d. across goods and consumers have the same search cost for all goods, then consumers will search in order of expected utility (see example 1). Still, there are at least three reasons this argument might fail: first, there may be more uncertainty about the hidden attribute for some goods than others, and this might lead individuals to search such goods first. Second, unobservables might be correlated across goods, so that, for example, learning good news about good 1 might cause one to positively update about good 2 and choose to search it before good 3 even if $\mathrm{EU}_{i3} > \mathrm{EU}_{i2}$. Third, search costs might vary across goods, meaning that consumers prefer to search goods with lower search costs first even if they exhibit lower expected utility than other products.

While the restriction that priors be i.i.d. and search costs be constant across goods is sufficient for assumption 2(i), it is not necessary. Indeed, our baseline result with $\gamma_1 \neq 0$ allows for non-i.i.d. priors about the unobserved attribute z. Alternatively, priors may be heterogeneous but consumers may be unsophisticated and fail to take into account option value, as in the directed cognition model studied in Gabaix et al. (2006). Consumers searching for a laptop online may enter some attributes into a search function and look at the items that rank highly according to those attributes without regard for whether a lower item is worth searching first because its value is more uncertain despite its lower average utility. Such examples also raise the natural concern that in many settings, factors such as the order in which items appear in search may impact search costs separately from expected utility. Applications in the marketing literature often allow search costs to vary with observable attributes, such as the position of a good in search (e.g., Ursu 2018). As discussed in section III.B, we extend our main result to allow for these violations of our expected utility assumption by considering cases where some observable attributes impact search but not utility.

Our second assumption on search is that consumers search good j if and only if $g_i(x_j, \epsilon_{ij}, \bar{u}) \geq 0$, where $\bar{u}$ is utility in hand; we also impose the natural restriction that one is (weakly) less likely to search as $\bar{u}$ increases. This assumption is satisfied in most search models we are aware of in the literature, including Weitzman search, satisficing, simultaneously searching all goods with expected utility above a threshold, random search, and directed cognition. One exception is a model in which consumers search simultaneously as in Chade and Smith (2006). This model would violate the assumption because the function g_i that determines whether i searches good j cannot be written only as a function of x_j and ϵ_{ij} since it will depend on the expected utility of all goods. In appendix section A.7, we show that our methods can be extended to accommodate one version of this model based on Honka, Hortaçsu, and Vitorino (2017). We also investigate the robustness of our approach to a violation of this assumption in the simulations of section IV of the supplementary appendix.

Our third assumption, that consumers choose the good that maximizes utility among searched goods, embeds two separate ideas. The first is that consumers do not choose a good they have not searched; this is natural in contexts such as e-commerce, where consumers typically have to open a product's page to add it to their carts. The second restriction is that consumers maximize utility given the information available. This could be relaxed by specifying a positive utility function that allows for consumer errors; as long as consumers maximize that positive utility function, the weight that they would attach to the hidden attribute given full information will be revealed. It is then up to the researcher whether to take this weight as the normative benchmark or whether to use some external standard.

The fourth assumption again nests two pieces: (i) that only the value of z_j is unknown before search, which we relaxed in section III.C, and (ii) that search reveals all information about the hidden attribute. This assumption is natural in settings where z_j is fully observed to the econometrician, as in our case, but is not always plausible. For instance, if the hidden attribute is school value added, a consumer who searches more may learn about test scores and graduation rates, but these are (imperfect) signals of the underlying variable. There is a literature on consumer (Bayesian) learning, which models more explicitly the case when search is not fully informative (see Erdem and Keane 1996; Ackerberg 2003; Crawford and Shum 2005; Ursu, Wang, and Chintagunta 2020, among others).

While our assumptions are not without bite, they subsume a range of search protocols and thus are less restrictive than fully specifying a structural search model. Still, one might wonder whether they will hold in empirically relevant settings. We address this in two ways. First, in section IV.C we show that the assumptions on the search process can be tested based on the same data required for estimation of the model. Second, we apply our approach to data from a lab experiment (sec. VII), where we are able to successfully recover preferences based on data with imperfect information, as well as observational data from Expedia (sec. VIII), where our testing approach correctly identifies the attribute that is not immediately visible to consumers.

B. *What Type of Data Is Required?*

The key input to our approach is a flexible estimate of s_1, the choice probability for the good with the highest $\tilde{z}$ as a function of the x and z attributes of all the products in the choice set. As in any discrete choice model, this requires enough variation in the x and z attributes across choice instances.

Standard full-information arguments assume that s_1 can be estimated at a choice set $(\mathbf{x}, \mathbf{z})$ and at a choice set that is identical except that z_j is perturbed by a small amount for some good j. Comparing these two markets nonparametrically pins down the first derivatives of the choice probability functions with respect to z_j. Similarly, nonparametric identification of $\partial s_1/\partial x_1$ requires a third market where x_1 varies holding fixed all other attributes. Thus, recovering $(\partial s_1/\partial z_1)/(\partial s_1/\partial x_1)$ requires at least three markets. The required variation in full-information models can come from individual or aggregate data and can come from cross-sectional variation across markets or panel variation within markets over time. When products have fixed attributes, one typically makes an exchangeability assumption—that is, that the distribution of the ϵ_{ij} shocks is the same regardless of the identity of product j; variation in product availability across markets then generates the desired variation in $(\mathbf{x}, \mathbf{z})$ even if those attributes are fixed for any given good. The exchangeability assumption likewise accommodates the case where the identity of good 1 (i.e., which good has the largest value of $\tilde{z}$) varies across markets.

In our approach, all of the above still applies—the only difference is that we need more data than under full information since we require second derivatives rather than first derivatives. Specifically, because we take derivatives with respect to three arguments, we need at least six markets to compute our key ratio.[15] Of course, in practice we typically do not observe such clean variation—usually, multiple elements of ($\mathbf{x}$, $\mathbf{z}$) move simultaneously. Just like in full-information models, this is where parametric restrictions on the choice probability functions are helpful, and data from additional markets or time periods can reduce the dependence on specific parametric assumptions.

One exception to the argument above is the case with endogenous product attributes (sec. III.A). In this setting, just like in full-information models, one needs data from many markets to recover the market-level unobservables that are correlated with the endogenous attributes (for the case with aggregate data, see Berry and Haile 2014; for the case with individual-level data, see Berry and Haile 2020).

C. *Testing Search Model Assumptions with and without Observable Search*

Our analysis so far has proceeded as if search were not observed; that is, we observe final choices as a function of $\mathbf{x}$ and $\mathbf{z}$, but we do not observe which specific goods were searched. Datasets increasingly contain some information on what is searched; for example, in online clickstream data, one observes not only which product was purchased but also which products were clicked en route to purchase (e.g., Ursu 2018). In many settings, it is plausible to assume that such clicks reveal which products were searched.

Can preferences be identified without resorting to our approach or an explicit search model in these cases? One might speculate that our identification results would be unnecessary; given data on which products were searched, perhaps preferences can be estimated by applying standard methods to the set of searched products without any of the assumptions we require here. However, this is not generally the case because the unobservable component of utility may also drive the search decision. In such cases, goods with undesirable observables that are searched likely have an especially high realization of ϵ. Thus, it may appear from choice probabilities conditional on search as though the observable attributes are not so bad when in practice, individuals dislike those attributes but this dislike is offset by a large ϵ.[16]

[15] Given a baseline market, recovering $\partial^2 s_1/\partial z_1 \partial z_j$ requires perturbing z_1 alone, z_j alone, and z_1 and z_j jointly. Similarly, recovering $\partial^2 s_1/\partial z_1 \partial x_j$ requires perturbing z_1 alone, x_j alone, and z_1 and x_j jointly. Thus, we need five markets plus the baseline market.

[16] A second reason unobservable components of utility might impact search is if preferences are unobservably heterogeneous (random coefficients). Even if search does not depend on ϵ, preferences cannot generally be recovered using only conditional choices unless IIA is satisfied. To see why heterogeneous preferences create a problem, imagine that products have quality ratings from 1 to 5. There are two types of consumers, one that cares about quality and one that does not. The type that cares about quality is indifferent about quality over the 4–5 range but values quality over the 1–4 range sufficiently that quality differences

Once our approach is used to identify preferences, clickstream data can be used to conduct additional overidentifying tests. For example, one can compare the order in which products are searched in the data with the order implied by the estimated model.

Even when we do not observe auxiliary information on which goods are searched, the assumptions in our model can be jointly tested by checking whether the observed choice probabilities are consistent with bounds implied by the estimated preferences and assumed search rule. To construct an upper bound on choice probabilities, note that a good j cannot be chosen if there is an alternative good with higher expected utility and higher utility. Thus, we have

$$s_j(\mathbf{x}, \mathbf{z}) \leq 1 - P(U_{ik} \geq U_{ij} \text{ and } \mathrm{EU}_{ik} \geq \mathrm{EU}_{ij} \text{ for some } k \neq j). \tag{9}$$

The latter probability can be directly computed from knowledge of preferences and the distribution of ϵ. To construct a lower bound, note that if good j maximizes both utility and expected utility, then it will be chosen. This implies that

$$s_j(\mathbf{x}, \mathbf{z}) \geq P(U_{ij} \geq U_{ik} \text{ and } \mathrm{EU}_{ij} \geq \mathrm{EU}_{ik} \text{ for all } k). \tag{10}$$

Once again, the probability on the right can be computed given knowledge of preferences and the distribution of ϵ. Checking whether our estimated choice probabilities are consistent with these bounds provides a test of the model.

Finally, our model is overidentified. For example, in our baseline case, $(\partial^2 s_1/\partial z_1 \partial z_j)/(\partial^2 s_1/\partial z_1 \partial x_j) = \beta$ for all alternative goods $j \neq 1$ and values of $(\mathbf{x}, \mathbf{z})$ at which the derivative in the denominator is nonzero. This provides a number of overidentifying restrictions that could be used to further test the model.

D. Which Counterfactuals Can Our Approach Address?

Here we discuss the class of counterfactual questions that can be addressed using our method. We start from applications that do not require recovering the distribution of search costs.

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outweigh any other differences observable to consumers. Suppose that quality is observable to consumers (x) but price is observed only conditional on search (z). Quality-conscious consumers only search goods with quality of at least 4. Other consumers will search all goods. If we estimate preferences conditional on search, we will wrongly conclude that no one cares about quality: quality conscious consumers do not care about quality given the goods they have searched (quality ranging from 4 to 5) and non-quality-conscious consumers do not care about quality at all. To estimate preferences correctly, we would have to jointly model the decision of which goods to search and preferences conditional on searching. Thus, with heterogeneous preferences, the existing literature requires specifying a search model to estimate preferences even when search is observed in the data. Our approach avoids the need to do this under the assumptions we have outlined. We relegate this point to a footnote because the estimation of high-order derivatives required to identify random coefficients models in our approach may often be impractical.

Full-information counterfactuals.—One important class of counterfactuals asks, How would consumers choose if search costs were reduced? The most natural counterfactuals in our baseline case involve directly informing consumers about the hidden attribute. These counterfactuals are natural in our setting because the hidden attribute is observable to the econometrician.[17] In these cases, knowing preferences is sufficient to simulate how information would impact choices without a structural search model, as we demonstrate in our lab and field experiments. In settings such as Hastings and Tejeda-Ashton (2008) or Allcott and Taubinsky (2015), where experimenters fully inform consumers about attributes of goods that were previously accessible at a financial or cognitive cost, our approach can be used to forecast the impact of interventions before they are conducted. Additionally, in section VII of the supplementary appendix we show how to quantify the welfare gains from more informed choices with an additional separability assumption. Of course, since our approach does not commit to a specific model of search, it does not speak to the gains directly stemming from reduced search costs. In this sense, the estimated increase in welfare can be viewed as a lower bound on the total gains from an information intervention. Estimating the reduction in search costs requires either fully specifying a search model and recovering the cost distribution (we discuss this more at the end of this subsection) or using some auxiliary data on, for example, time spent searching and value of time.

Advertising and product design.—As a second related example, consider a firm trying to understand which features to emphasize in the advertising of a product. Conditional on visible attributes, our results could be used to identify features that consumers value but are not currently always aware of. The firm could use this insight to optimize its advertising strategy, as well as to inform the design of new products (see, e.g., Becker and Murphy 1993; Bagwell 2007).

Normative evaluation of choices.—In many counterfactuals where limited information or search costs are not the primary object of interest, one nonetheless is concerned about accurately valuing the attributes of goods. An example is a subsidy for environmentally friendly automobiles. To evaluate such a subsidy, one would conventionally estimate demand and cost parameters in the automobile market (Berry, Levinsohn, and Pakes 1995). If the market were otherwise competitive and efficient, the subsidy might distort choices (creating deadweight loss) but have offsetting externalities. If, however, some consumers are unaware of differences in energy

[17] This can be contrasted with cases where information is only partial and so some search costs likely remain. For example, when unobservables are revealed by search (as in sec. III.C), some information that consumers learn upon search is not observable to the econometrician, so informing consumers about the observable component would not eliminate the need to search.

efficiency, the subsidy might redirect them to the cleaner products they would value most if they had more information, meaning that it may be both privately and socially desirable. Our methods can be used to recover whether, before imposing the subsidy, consumers are informed about differences in energy efficiency.

We have focused so far on applications where search costs do not need to be recovered. However, our model can also be used to identify search costs given preferences and an underlying structural search model. In section VI of the supplementary appendix, we give an explicit example of how search costs can be recovered in a Weitzman model once preferences are known. Intuitively, when preferences are known we know how consumers would respond to the hidden attribute with zero search costs, and thus we can trace out the distribution of search costs from the observed responsiveness of choice probabilities to the hidden attribute. In section VI of the supplementary appendix, we also discuss in more detail questions that can be addressed once search costs are recovered.

V. Estimation

Our identification results show that preferences can be recovered given knowledge of the choice probability function for good 1, denoted by $s_1(\mathbf{x}, \mathbf{z})$. We now discuss how s_1 can be estimated from data on choices and product attributes. We focus on the case with individual-level data (as in the experiment of sec. VII and the application of sec. VIII). However, our identification approach also applies to aggregate (i.e., market share) data as long as one can consistently estimate the choice probability functions s_j, as discussed in section IV.B. With individual-level data, our setup implies the following conditional moment restrictions:

$$E(y_j - s_j(\mathbf{x}, \mathbf{z})|\mathbf{x}, \mathbf{z}) = 0 \quad \forall j, \tag{11}$$

where y_j is a dummy variable equal to one if a consumer chooses good j. Thus, methods designed to estimate conditional moment restriction models can be used. Of course, the performance of an estimator will depend on how flexibly it captures the derivatives that identify preferences in our approach.

We consider two approaches to estimating $s_1(\mathbf{x}, \mathbf{z})$ and thus its derivatives: (i) an approximation via Bernstein polynomials, which is viable when the number of goods and attributes is small, and (ii) a "flexible logit" model, which is more ad hoc but scales better as the number of goods increases.

Both estimation approaches involve choosing tuning parameters; for example, in the Bernstein polynomial approach one must choose the degree of the polynomial approximation. To deal with this problem, in our experimental analysis we preregistered the code used to analyze the experiment.

We would encourage other researchers using our method to do the same to avoid concerns about post hoc tuning to obtain desired results.

A. *Approximation via Bernstein Polynomials*

Following Compiani (2022), one can approximate the demand function via Bernstein polynomials. This allows the researcher to impose natural restrictions via linear (and thus easy to enforce) constraints on the coefficients to be estimated. Specifically, the class of models considered in this paper satisfies standard monotonicity restrictions in $\mathbf{x}$ and $\mathbf{z}$ (s_j increasing in x_j and z_j and decreasing in $\boldsymbol{x}_{-j}$ and $\boldsymbol{z}_{-j}$). In addition, one can consider other constraints, such as exchangeability across goods, which requires demand to depend only on the attributes of the goods, not their identity.[18] Exchangeability is satisfied if the unobservables entering demand (e.g., preference parameters and shocks as well as search costs) have the same distribution across goods. We impose both monotonicity and exchangeability in the Bernstein polynomial results reported below. The purpose of these restrictions is twofold. First, they discipline the estimation routine in the sense that they help obtain reasonable estimates of quantities of interest (e.g., negative price elasticities). Second, they help partially alleviate the curse of dimensionality that arises as the number of goods increases. The coefficients in the Bernstein approximation of s_j can be estimated by minimizing a generalized method of moments objective function based on the restrictions in (11) subject to the constraints. In section IV of the supplementary appendix, we report results from numerous simulations with a variety of data-generating processes that suggest that this estimation approach performs well with a small number of goods (two or three) when the assumptions of our model are satisfied. It also consistently outperforms standard logit estimates in simulations where our assumptions are violated. This approach has the advantage of providing a nonparametric approximation to the choice probability function s_1 but it suffers from a curse of dimensionality that makes it very data demanding for four or more goods.

B. *"Flexible Logit"*

As the number of goods increases, nonparametric methods face a curse of dimensionality and thus it becomes necessary to place some additional structure on the problem. In this section, we develop one such approximation that performs well in simulations for a larger number of goods.[19]

[18] For a formal definition of exchangeability, see Compiani (2022).

[19] For example, in our application in sec. VIII, we have 10 products per choice set and flexible logit involves 46 parameters in total. In contrast, a fully nonparametric approach

As discussed in more detail in section V of the supplementary appendix, conventional full-information models typically impose strong restrictions on the structure of the derivatives of choice probabilities. For example, in a multinomial logit model, it is always the case that $(\partial^2 s_1/\partial z_1 \partial z_j)/(\partial^2 s_1/\partial z_1 \partial x_j) = (\partial s_1/\partial z_j)/(\partial s_1/\partial x_j)$. Since our test for full information is based on detecting discrepancies between these two ratios, a logit model will clearly not be suitable. To allow for the required additional flexibility, we let the mean utility for good 1 depend directly on attributes of rival goods as follows:

$$ax_1 + b_1 z_1 + \sum_{k\neq 1} (\eta_k w_{z1k} z_k + \eta_{2k} w_{x1k} x_k + \rho_k w_{z2k} z_k z_1 + \rho_{2k} w_{x2k} x_k z_1), \qquad (12)$$

where w_{z1k}, w_{x1k}, w_{z2k}, and w_{x2k} are known weights and a, b_1, η_k, η_{2k}, ρ_k, and ρ_{2k} are coefficients to be estimated. Further, we let the mean utility for $k \neq 1$ take the standard form $ax_k + bz_k$. If we set the weights equal to one, this would be a conventional logit model where the utility for good 1 is allowed to depend flexibly on the attributes of good 1, as well as rival attributes interacted with the attributes of good 1. This is the idea behind the "universal logit" model (McFadden 1981, 1984), and thus our method can be viewed as an extension of this designed to target our second derivatives of interest. Specifically, approximating choice probabilities with estimates from a standard logit model, it is possible to choose the weights as a function of choice probabilities so that the structural parameter of interest (β/α) is a closed-form function of the estimated coefficients (we show this formally in sec. V of the supplementary appendix). More precisely, with weights chosen appropriately,

$$\frac{\partial^2 s_1}{\partial z_1 \partial z_j} \Big/ \frac{\partial^2 s_1}{\partial z_1 \partial x_j} = \frac{-b + \eta_j + \rho_j}{-a + \eta_{2j} + \rho_{2j}}, \qquad (13)$$

where the coefficients on the right-hand side are recovered directly by estimating the model based on equation (12). The weights are estimated as functions of choice probabilities and coefficients from a naive logit model.[20]

We note that the parameters in (12) do not have a causal interpretation (i.e., we are not positing that the actual utility of good 1 depends on the

would involve a much larger number of parameters (e.g., even if we modeled s_1 as a polynomial of up to degree 1 in each of the arguments $(\mathbf{x}, \mathbf{z})$, we would have $2^{20} = 1{,}048{,}576$ parameters).

[20] Specifically, we assume that $u_{ij} = \alpha^* x_j + \beta^* z_j + \epsilon_{ij}$ and compute the implied choice probabilities s_j^*. Then, $w_{z1j} = w_{x1j} = s_j^*/(1 - s_1^*)$ and $w_{x2j} = w_{z2j} = \beta^*(((1 - 2s_1^*)s_j^*)/(1 - s_1^*))(1 + \beta^*(1 - 2s_1^*)z_1)^{-1}$.

attributes of good k for $k \neq 1$). Instead, (12) is simply a flexible function of $(\mathbf{x}, \mathbf{z})$ that we found captures the second derivatives of s_1 well. In section IV of the supplementary appendix, we show that the flexible logit performs extremely well in simulations for a variety of data-generating processes. For three data-generating processes satisfying the assumptions of our model, conventional logit estimates are biased, but flexible logit confidence intervals include the true values. For a fourth data-generating process violating the assumptions of our model, flexible logit has a small bias with a large number of goods but is consistently less biased than the standard logit estimates. This being said, the approach is somewhat ad hoc, and we welcome more formally validated approaches for optimizing the trade-off between flexibility and scalability in estimating our second derivatives of interest.

In subsequent applications, we have found that the performance of flexible logit deteriorates when the impact of z on choices is small in the data, leading confidence intervals to explode. This is intuitive regardless of how our model is estimated: if $\partial s_1/\partial z_1$ is small, then $(\partial^2 s_1/\partial z_1 \partial z_j)/(\partial^2 s_1/\partial z_1 \partial x_j)$ is difficult to estimate with any precision unless the dataset is extremely large. Our approach is thus better suited to settings where a potentially hidden attribute has a meaningful impact on choices in the raw data (e.g., school choice as a function of high school test scores).

VI. Practitioner's Guide

Table 1 provides guidance for applied researchers seeking to use our approach. As in the above section, we focus on the case with individual-level data, although these procedures could be adapted to the case with aggregate market-level data. We also assume the researcher is willing to fix the variance of ϵ via parametric assumptions on its distribution (e.g., assume it is Gumbel distributed), so that α cannot be normalized without loss and instead needs to be estimated. Further, we focus on the case with continuous attributes and discuss how to handle discrete attributes in appendix section A.8. Section I.2 of the supplementary appendix outlines how to adapt the approach to the case where the researcher wishes to let x enter utility nonlinearly.

The estimation routines outlined in table 1 are likely to lead to slower-than-parametric convergence rates. This is because our parameters of interest are estimated by evaluating nonparametric functions at specific points and/or using a slice of the data (e.g., for the estimation of α in step 2[c]).[21] However, we are reassured by the fact that the methods perform well in both our experiment and our empirical application.

[21] This step is akin to estimation for varying coefficients models (Cai, Fan, and Li 2000).

TABLE 1
STEPS FOR ESTIMATION

1) For each choice set in the data, identify good 1: the good with the highest $\tilde{z}$ (or, if z is vector valued, the good with the highest weighted $\tilde{z}$-index obtained via the procedure in app. sec. A.3).
2) For the baseline model of section II.B, follow these steps:
   a) If the number of goods is two or three, estimate the model nonparametrically:
      i) Use equation (11) to estimate the choice probability function s_1 using Bernstein polynomials subject to monotonicity and exchangeability restrictions (Compiani 2022).
      ii) Compute estimates of $\partial^2 s_1(\mathbf{x},\mathbf{z})/\partial z_1 \partial z_j$ and $\partial^2 s_1(\mathbf{x},\mathbf{z})/\partial z_1 \partial x_j$ for all $j \neq 1$ and all choice sets $(\mathbf{x},\mathbf{z})$ in the data.
      iii) Take the ratio of a trimmed mean of $\partial^2 s_1(\mathbf{x},\mathbf{z})/\partial z_1 \partial z_j$ to a trimmed mean of $\partial^2 s_1(\mathbf{x},\mathbf{z})/\partial z_1 \partial x_j$ (both means are over choice sets $(\mathbf{x},\mathbf{z})$). Further average these ratios over $j \neq 1$ to obtain an estimate of β/α.
   b) If the number of goods is greater than three, estimate the model using flexible logit (sec. V.B):
      i) Create initial naive estimates (α^*, β^*) and associated choice probabilities s_j^* by estimating a naive logit model. Use these to construct the weights w_{ij} as described in section V of the online appendix.
      ii) Estimate equation (12) given these weights to recover the coefficients: a, b, $(\eta_k, \eta_{2k}, \rho_k, \rho_{2k})_{k \neq 1}$.
      iii) Use equation (13) to compute $(\partial^2 s_1/\partial z_1 \partial z_j)/(\partial^2 s_1/\partial z_1 \partial x_j)$, and take an average across j (or impose coefficient restrictions such that η_j, η_{2j}, ρ_j, ρ_{2j} are the same for all j). This gives an estimate of β/α.
   c) Recover β as follows:
      i) Estimate $\alpha + \beta\gamma_1$ via standard full-information models (e.g., logit), using only choice sets where the variance of $\tilde{z}_j$ across j is below a cut-off.
      ii) Given the estimate of β/α from step 2(a) or 2(b) and the estimate of $\alpha + \beta\gamma_1$ from step 2(c)(i), solve for α and β (given known γ_1).
3) If some attributes are endogenous, modify step 2(a)(i) to incorporate instruments (as in Compiani 2022). Then follow steps 2(a)(ii) and 2(a)(iii) where now the choice probability s_1 is a function of not only $(\mathbf{x},\mathbf{z})$ but also the unobservables causing endogeneity ξ (which can be fixed once the structural demand function is estimated).
4) If the data contain variables that are likely to affect search but not utility (e.g., rankings r on a web page), drop all choice sets where consumers choose a product ranked lower than product 1 $(r_j < r_1)$. Then, follow step 2 using the set $\mathcal{R} = \{j : r_j \geq r_1\}$ as the choice set and including r among the x variables. In step 2(c)(i), also condition on choice sets where the variance of r_j across j is sufficiently low.
5) To allow for the possibility that ϵ is revealed only via search, choose product j in step 2(a)(ii) or 2(b)(iii) to be the good with the best value of x (which could coincide with good 1).
6) Optional:
   a) To increase power, repeat the previous steps for various choices of the x variable (if available) and take a weighted average of the resulting β estimates with weights proportional to their precision.
   b) Test the model by checking whether the estimated choice probabilities lie within the bounds derived in section IV.C.

NOTE.—We recommend using simulations to choose the following tuning parameters before estimation: the polynomial degree in step 2(a)(i), the amount of trimming in step 2(b)(iii), and the variance cutoffs in steps 2(c)(i) and 4.

VII. Experimental Validation

Our identification proof and simulation results show that preferences can be estimated regardless of whether consumers are fully informed, provided consumers search in a way that is consistent with our assumptions. Of course, this does not tell us whether those assumptions are likely to be satisfied in practice.

In this section, we test in a lab experiment whether we can recover preferences in a setting where consumers engage in costly search. Unlike in our simulations, the search protocol is unknown to us and not restricted to satisfy all of our model assumptions. In particular, while we enforce the assumption that consumers must search a product to choose it, we do not constrain the order in which the various options are searched or the rule that determines when to stop searching. We nonetheless show that we are able to correctly recover preferences using our "search-robust" estimation technique.

A. Setup

We selected 1,000 books for sale on Amazon Kindle chosen from a wide variety of genres. For each book, we observe its average rating on Goodreads.com as well as the average rating from Amazon.com, the number of reviews on Goodreads, and the price of the book for Amazon Kindle.

In our experiment, conducted via Connect CloudResearch,[22] each participant made 20 choices from sets of three randomly selected books. In each choice, participants had to select a book—that is, they had no outside option. For all books, participants could see a photo of the cover; the title, author, and genre; and the Goodreads rating and total number of ratings. Prices were randomized to integers from \$11 to \$15 (equally likely). All books were then further discounted by an integer amount uniformly drawn from \$0 to \$10 (and participants were informed of this). All users were given a \$15 bank at the start of each choice, from which any costs incurred were deducted. There were a total of 196 participants, yielding 3,920 choices.

The discount is our key variable of interest. For five of the 20 choices, users could see all discounts and thus could see the net price of all options at no cost. For 15 of the 20 choices, discounts were hidden and users had to pay a cost to see the discount for any given book.[23] The cost per click was constant for each user across the 15 choices and randomly chosen from

[22] We preregistered the experiment and estimation codes at the platform Open Science Framework: https://doi.org/10.17605/OSF.IO/D4XA2.

[23] The full-information and costly-information choice situations were randomly ordered, so that the five full-information choices were intermixed with the costly-information choices.

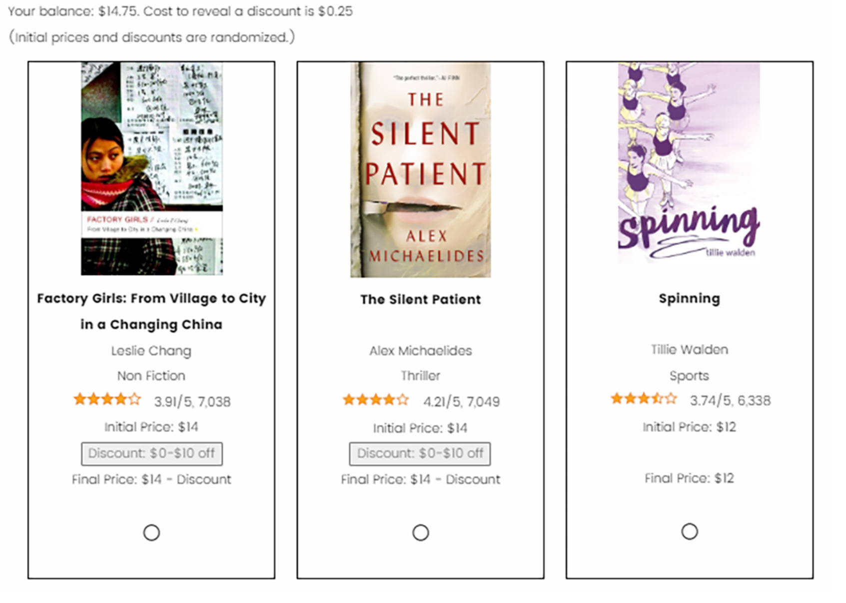

FIG. 1.—Lab experiment: sample product selection screen.

{$0.10, $0.25, $0.35, $0.50}. For the 15 choices with hidden information, users could choose books only after they clicked to reveal the discount and had to choose at least one book. One of the 20 choices made by each user was randomly chosen to be realized, and users received the chosen book as well as any money left over from the original $15.

Figure 1 shows a sample product selection screen from a choice where discounts were hidden. In this case, the user clicked to reveal the discount of the second book and could either choose that book or continue by revealing the discounts for additional books. Note that the user could search books in any order she wished. The average total number of clicks per participant per choice is 1.5.

The five choices where all information is revealed are our benchmark for the "truth." The goal is then to test whether the weight on discounts relative to prices that we estimate in the cases where discounts are costly to observe matches the relative weight we see when discounts are visible to everyone (i.e., can we predict informed choices using data from uninformed ones). Further, because both discounts and prices are in dollar terms and because they are randomized (and thus not signals of quality), there is a second benchmark: if consumers are rational, the weight on discounts and prices should be equal.

We model choices using the following utility specification:

$$U_{ij} = price_j \cdot \alpha_1 + rating_j \cdot \alpha_2 - discount_j \cdot \beta + \epsilon_{ij}, \tag{14}$$

where ϵ_{ij} is i.i.d. type I extreme value and accounts for any aspect of consumers' tastes for books (based on the title, image, author, or genre) not summarized by the price, discount, and rating variables. Fully informed and rational consumers should have $\alpha_1 = \beta$. Our goal is to show that we can recover these fully informed preferences using the choices of participants for whom revealing discounts is costly.[24]

B. *Estimation Results*

Columns 1 and 2 of table 2 show results from estimating a standard logit model on consumer choices for the five choice situations per consumer where all information is revealed (Full Info) and the 15 choice situations where consumers must pay to reveal information (Costly Info), respectively. With full information, consumers place equal weight on prices and (negative) discounts, so they pass our test of rationality. In other words, they care only about the final price of the product. By contrast, when discounts are costly to reveal, the coefficient on the discount variable in the standard logit model is attenuated (the "Costly Info" column). This is because consumers are insensitive to variation in discounts for books they do not search. The ratio of the two coefficients is 1.094 in the full-information treatment and 0.665 in the costly-information treatment, consistent with the attenuation bias established in lemma 3.

Following section V.A, we estimate the demand function $s_1(\mathbf{x}, \mathbf{z})$ via Bernstein polynomials. The exact procedure is described in section VIII of the supplementary appendix. Our estimate of β/α_1 is 1.185, and the confidence interval contains the corresponding estimate of 1.094 from column 1 (and the theoretical estimate of 1) and is sufficiently tight to exclude the standard logit estimates in the costly-information treatment.[25] Besides estimating β/α_1, we need to directly recover the α coefficients. Consistent with the proof of lemma 2, we compute these by estimating a logit model using only choice sets where the variance of the discount across goods is in the bottom quartile. The results are reported in column 3 of table 2, along with the value of β implied by our estimates of α_1 and β/α_1. Again, the confidence intervals include the full-information

[24] Note that, given our parametric assumption on the distribution of ϵ, we are not free to normalize any of the α coefficients here.

[25] Consistent with the guide in table 1 and the preregistered code, we calculate β/α_1 as the ratio of trimmed means of suitable second derivatives, where the averaging is over choice sets. At the population level, one could equivalently invert the order and take the (trimmed) mean of the ratios, but in simulations we found that this estimator tends to be biased in finite samples. In our experiment, it is equal to 0.73.

TABLE 2
STANDARD LOGIT AND SECOND DERIVATIVE ESTIMATION RESULTS

	Standard Approach		Our Approach
Variable	Full Info (1)	Costly Info (2)	Costly Info (3)
Price	−.224***	−.182***	−.230***
	(.031)	(.017)	(.032)
Discount (−)	−.245***	−.121***	−.273***
	(.015)	(.008)	(.057)
Rating	.350**	.471***	.588***
	(.166)	(.091)	(.171)
Discount (−)/price	1.094***	.665***	1.185***
	(.162)	(.077)	(.216)
Observations	980	2,940	2,940

NOTE.—This table shows estimation results from a standard logit model estimated on the full-information and costly-information treatments in cols. 1 and 2 and Bernstein polynomials estimation of the second-derivative ratio on the costly-information treatment in col. 3. The minus sign indicates that discount is multiplied by −1 so that the coefficient on discount should equal that of the price under full information. Standard errors on the discount coefficient and the ratio of the discount and price coefficients are computed using 1,000 bootstrap draws.
** Significant at the 5% level.
*** Significant at the 1% level.

values. In other words, using data only on choices when information is costly, we successfully recover informed preferences.[26]

Having recovered all preference parameters, we can compute how information will change behavior and choice quality. Using data on choices only when search is costly, our model predicts that on average full-information consumers would save $0.71 per choice from choosing books with lower discounts. The corresponding number in the data is $0.64 per choice situation, since consumers in the costly-information treatment average discounts of $5.82, while consumers in the full-information treatment average discounts of $6.46.[27] In other words, we can accurately predict how consumers will respond to information provision before the information is provided. We can also compute the dollar-equivalent welfare benefits of providing consumers with information. To do so, we take our estimates from column 1 as the normative preferences (i.e., as the correct metric to compute consumer welfare) and calculate by how much welfare changes when consumers go from making partially uninformed choices to fully informed choices using the approach in section VII of the supplementary

[26] To show validity of the nonparametric bootstrap for the case where the constraints are not binding asymptotically, one can combine the general framework of conditional moment models with different conditioning variables (Ai and Chen 2007) with the results in sec. V of Chen and Pouzo (2015). An interesting avenue for future research is to explore the case where the constraints are binding asymptotically.

[27] We look at differences in discounts as opposed to final prices paid since the latter are essentially the same in the full-information and in the costly-information conditions.

appendix. We then repeat this exercise using the estimates from column 3 as the normative preferences. We estimate an average welfare gain of \$0.18 per choice based on column 1 and of \$0.26 based on column 3. Thus, our model again yields results that are fairly close to those coming from the "true" fully informed choices in the data.[28]

As in most real-world settings, expected utility is not observable to the econometrician in our experiment: while we can see attributes of the goods in question, we do not know how individuals will weigh these attributes, nor do we know their preferences for specific genres or book titles and images (captured by ϵ_{ij} in our model). The assumption that consumers search in descending order of expected utility is substantive and could be violated in numerous ways: users might always reveal discounts for the lowest-priced book first or they might search in the order in which books are displayed. Nonetheless, our "robust" estimation approach succeeds in recovering the preferences that consumers reveal in the full-information condition. In section IX of the supplementary appendix, we report results from the test discussed in section IV.C, showing additionally that the estimated choice probabilities lie within the upper and lower bounds implied by our expected utility assumption. Thus, we fail to reject the assumption.

VIII. Field Validation

Our lab experiment demonstrates one setting where our approach correctly identifies hidden attributes and forecasts how consumers will respond to information about hidden attributes. Of course, this leaves open the question of whether we can identify preferences in real-world settings with a larger number of goods, where search costs are implicit and potentially heterogeneous and where we (as experimenters) cannot strictly control the information available to consumers.

In this section, we investigate these issues using publicly available data from a leading online travel agency, Expedia (Ursu 2018).[29] The data we use include transactions from 54,648 consumers over an 8-month period between November 1, 2012, and June 30, 2013.[30] At the time, consumers would search for a hotel in a given city and Expedia would present a list of available options. On the list, consumers observe a range of hotel characteristics, including the price per night, whether the hotel is on promotion or part of a chain, the star rating, and the review score.

[28] Note that the benefits are smaller than the increase in discounts because information induces consumers to be more responsive to discounts, sacrificing some value on unobservable factors. Further, we put the word "true" in quotes because the full-information logit may be misspecified even in col. 1 due to, e.g., correlation in ϵ across products.

[29] The data are available for download at https://www.kaggle.com/c/expedia-personalized-sort/data.

[30] This is the final dataset. Section X.1 of the supplementary appendix contains details of data cleaning and variable descriptions.

TABLE 3
Summary Statistics

	Observations	Mean	Median	Standard Deviation	Minimum	Maximum
Price ($)	546,480	162.11	139.57	92.94	10	1,000
Stars	546,480	3.42	3.00	.91	0	5
Review score	546,480	4.01	4.00	.71	0	5
Chain	546,480	.64	1.00	.48	0	1
Location score	546,480	3.26	3.22	1.45	0	7
Promotion	546,480	.34	.00	.47	0	1

One attribute—location—is not visible to consumers in the search results and is visible only with additional effort, either clicking on the hotel or clicking to open a map showing unlabeled pins and then finding the pin of a hotel. The dataset contains a measure of location desirability, but this measure is not visible to consumers in the search results. We thus ask, Can our testing approach in lemma 4 correctly recover that location is a hidden attribute, whereas other attributes are directly visible on the search results page?

Note that while the expected utility assumption is not ex ante unreasonable in this setting, there are several ways that it might be violated. The most obvious is that the ranking of hotels in search results might matter directly, and we allow for this in an extension below. Consumers may ignore the search results and click on a map, where they can see location but not the other attributes of hotels (a very different search process). Or consumers might make errors, such as searching lower-priced hotels first with no consideration for how expected location varies with price. While we cannot observe these behaviors directly, we construct bounds on choice probabilities implied if the expected utility assumption is satisfied.

Table 3 provides summary statistics. Hotels on average charge $162 per night, with 3.5 stars and a review score of 4 out of 5. Sixty-four percent of the hotels belong to a chain, and 34% of the hotels display a promotion. Location attractiveness is a score ranging from 0 to 7 designed by Expedia to measure how centrally a hotel is located, what amenities surround it, and other aspects of location desirability. The average hotel has a location score of 3.26. Figure 2 illustrates how hotel characteristics appear

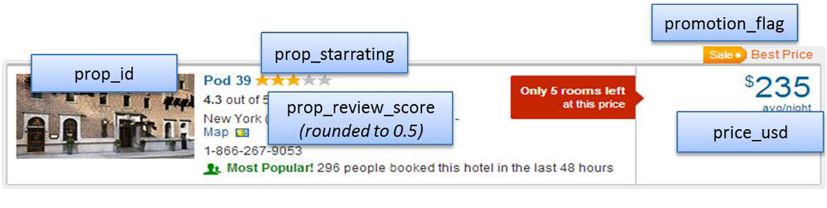

Fig. 2.—Hotel characteristics shown in the search impression.

TABLE 4
Model Specifications

Model	$\mathbf{x}_j$	z_j
I	Price, stars, review score	Location score
II	Stars, review score, location score	Price
III	Price, stars, location score	Review score
IV	Price, review score, location score	Stars

to consumers in search. Note that information on features such as price, stars, review score, and promotion flag are saliently displayed. However, consumers do not observe the detailed map unless they click, and the quantitative location score is not shown. To evaluate the attractiveness of a location, consumers need to spend time and effort to examine the map.

We consider choices with the following utility specification:

$$U_{ij} = \tilde{\mathbf{x}}_j \cdot \tilde{\alpha} + \mathbf{x}_j \cdot \alpha + z_j \cdot \beta + \epsilon_{ij}, \quad (15)$$

where ϵ_{ij} is type I extreme value and $\tilde{\mathbf{x}}_j$ is the vector of attributes that we always model as visible, including the chain, promotion, and position dummies.

For the four remaining variables—stars, review score, location score, and price—we estimate four models where one of these variables plays the role of z_j in our model (i.e., it may be revealed only after search), and the other three serve as $\mathbf{x}_j$ (i.e., they are assumed to be known before search). Table 4 shows the model specifications. We start by assuming that consumers do not form expectations on the hidden attribute based on the visible attributes and relax this assumption in a robustness check. We estimate each model using the "flexible logit" approach described in section V.B. Since we have several different x variables for each model, to increase power we estimate β by taking derivatives with respect to each individual x variable and then average the results (step 6 of table 1).[31]

Figure 3 shows the estimates from standard logit and flexible logit for each candidate z variable.[32] Location score is the only variable where we see clear evidence that standard logit is attenuated relative to flexible logit, which is consistent with the fact that location is not immediately available to consumers in the results page. Thus, standard logit tends to underestimate how much consumers value location. On the contrary, for the visible attributes—price, review score, and star rating—we find that the flexible logit confidence intervals include the standard logit estimates, and the differences between flexible logit and standard logit estimates are not significantly different from zero.

[31] Section X.2 of the supplementary appendix contains a step-by-step guide for computing the point estimates and confidence intervals.

[32] Values are reported in sec. X.3 of the supplementary appendix.

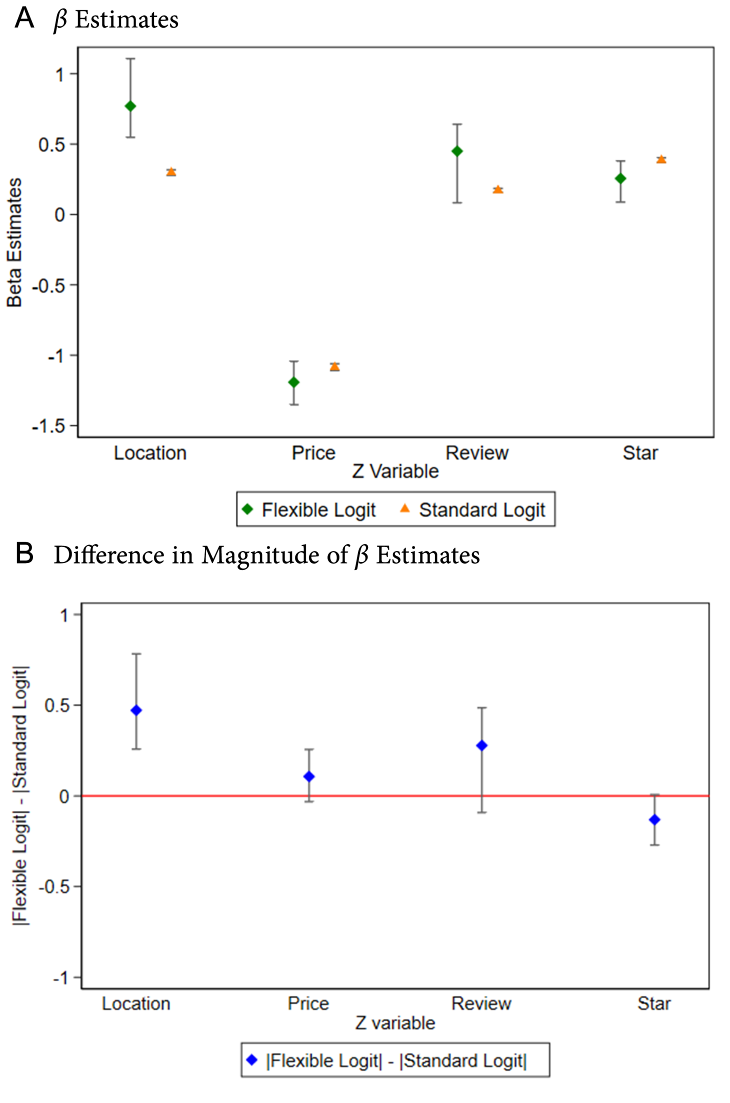

FIG. 3.—Estimation results. In *A*, we report 95% confidence intervals for the coefficient β for different choices of the z variable. In each case, we normalize the coefficient by multiplying it by the standard deviation of the variable. In *B*, we report 95% confidence intervals for the difference between the absolute value of the normalized β estimate from flexible logit and the absolute value of the corresponding standard logit estimate.

TABLE 5
Counterfactual Results

	Status Quo (1)	Counterfactual (2)
Average value for transacted item:		
Price ($)	143.52	137.06
Stars	3.47	3.42
Review score	4.03	4.03
Chain	.64	.64
Promotion	.40	.39
Position	4.55	4.78
Location score	3.32	3.41
Welfare difference per choice ($)		2.20

Note.—This table shows the average value of different attributes for the transacted item in the data (col. 1) and in the counterfactual scenario where consumers have full information on location (col. 2). The last row reports the average welfare change from the status quo to the counterfactual.

Given the evidence that location is a hidden attribute, we use our estimates of consumer preferences to compute how information about location will change behavior and choice quality. Table 5 shows the counterfactual results when we make the location score information visible to all consumers. The average location score among transacted hotels increases from 3.32 in the data to 3.41 in the counterfactual scenario where location is fully visible. Further, using the approach in section VII of the supplementary appendix, we compute the welfare benefits of providing consumers with location score information. We estimate an average welfare gain of $2.20 per choice, which is 1.5% of the average transaction price. Interestingly, the average price paid is higher in the status quo relative to the full-information counterfactual, suggesting that the platform is benefiting from consumers' imperfect information.

In section X.4 of the supplementary appendix, we conduct a series of additional analyses to check the robustness of the results. Specifically, we show that our results qualitatively do not change if (i) we allow consumers to form expectations about location based on the other attributes (i.e., $\gamma_1 \neq 0$), (ii) we let the position of the hotel on the results page affect the order of search as in section III.B, or (iii) we let the idiosyncratic term ϵ_{ij} be revealed to consumers only after search (sec. III.C). Additionally, (iv) we report results from the test discussed in section IV.C and show that the estimated choice probabilities almost always lie within the upper and lower bounds implied by the assumption that consumers search in descending order of expected utility. Further, (v) we check whether consumers who searched good 1 are systematically different from others by comparing the total number of clicks conditional on searching good 1, and we find that they do not behave differently from other consumers;

(vi) we explore robustness to how much data we use to estimate the coefficients on the visible attributes (which theoretically requires $\tilde{z}_j = \bar{z}$ for all j), and we find that the estimates from flexible logit are stable as we vary the amount of data and significantly different from the standard logit estimates; and (vii) we present a test of whether the estimates of the β coefficient on the location score vary across choices of which x variable is used to construct our ratio of second derivatives. We fail to reject the null hypothesis that the estimates obtained from different choices of x variables are equal.

IX. Conclusion

We give sufficient conditions to estimate preferences using only data on attributes and choices in cross-sectional or panel data even when consumers must search to acquire information about product attributes. The approach is robust in the sense that it works regardless of whether consumers have full information or engage in search based on a broad class of search protocols. Further, our results can be used to test whether consumers are fully or only partially informed about a given attribute.

Because our conditions allow preferences to be recovered when consumers are imperfectly informed, our results allow a wide range of inquiries that are impossible using conventional methods. First, before conducting an information intervention, choice data can be used to counterfactually estimate how consumers would choose were they fully informed. Further, our approach can be used to conduct reliable welfare analyses that are based on consumers' true preferences as opposed to the preference estimates obtained from standard methods assuming full information, which may be confounded by consumers' lack of information.

In many settings, our assumptions may not be exactly satisfied, but these must be assessed relative to the alternatives. The vast majority of empirical work currently makes the often dubious assumption that consumers are fully informed about all attributes of products.[33] Even if one lacks contextual information to support our assumption that consumers search in descending order of expected utility, our approach is much weaker than the standard assumption of full information and may be preferable in settings where estimates of preferences are needed to conduct welfare analysis.

Relative to methods based on the full-information assumption, the main downside of our approach is that it is more demanding of the data

[33] We count 350 articles published in the *American Economic Review*, *Quarterly Journal of Economics*, *Journal of Political Economy*, *Econometrica*, or *Review of Economic Studies* since 2015 that estimate discrete choice models. Of these 350, 315 (90%) assume that consumers are fully informed. The list of papers and their classification is available from the authors upon request.

and places limits on the extent of consumer heterogeneity allowed. For example, our identification arguments require that consumers agree about whether the possibly hidden attribute is good or bad, and further, recovering the distribution of preference heterogeneity requires parametric restrictions or very large datasets.

A second shortcoming of our approach currently is the ad hoc nature of the "flexible logit" estimation procedure that we use in large choice sets. We find that this functional form works well in a range of simulations, but we have not stated formal assumptions under which this estimation procedure will recover the true choice probability function. One implication of our theory is that the strong assumptions made in existing parametric models about cross derivatives rule out realistic behavior. Work clarifying the formal assumptions necessary to relax these restrictions while maintaining scalability in large choice sets would be welcome.

Our assumptions are sufficient for identification but not necessary, raising many questions: are there other conditions aside from the expected utility assumption that permit analogous data-driven identification of consumers who maximize utility? Are there necessary and sufficient conditions for preferences to be recoverable from choice data when consumers have partial information? Finally, our approach could be combined with consideration set methods to allow for imperfect information at both the attribute and the product level. This may be desirable to assess information interventions that inform about both the attributes of products and which alternatives exist.

Appendix

Additional Proofs

In this appendix, we collect the proofs not included in the main text. Throughout, we let $\mathcal{J} \equiv \{1, \ldots, J\}$ and often drop the i subscript for notational simplicity. Table A1 presents key notations used in the proofs.

TABLE A1
NOTATION

Notation	Definition	Section
P_4	$P(U_1 \geq U_k \; \forall \; k)$	A1
P_5^S	$P\,(\{U_1 \geq U_k \; \forall \; k\} \cap \{\mathrm{EU}_j \geq \mathrm{EU}_1$ for at least one $j \in \mathcal{S}\}$ $\cap \{g\,(x_1, \epsilon_1, U_j) \leq 0$ for all $j \in \mathcal{S}\} \cap \{g\,(x_1, \epsilon_1, U_j) \geq 0$ for all $j \in \mathcal{J}_{-1} \backslash \mathcal{S}\})$	A1
$P_{5,1}^S$	$P\,(\{U_1 \geq U_k \; \forall \; k\} \cap \{g\,(x_1, \epsilon_1, U_j) \leq 0$ for all $j \in \mathcal{S}\} \cap \{g\,(x_1, \epsilon_1, U_j) \geq 0$ for all $j \in \mathcal{J}_{-1} \backslash \mathcal{S}\})$	A1
$P_{5,2}^S$	$P(\{\mathrm{EU}_1 \geq \mathrm{EU}_j$ for all $j \in \mathcal{S}\} \cap \{g\,(x_1, \epsilon_1, U_j) \leq 0$ for all $j \in \mathcal{S}\}$ $\cap \{g\,(x_1, \epsilon_1, U_j) \geq 0$ for all $j \in \mathcal{J}_{-1} \backslash \mathcal{S}\})$	A1
$P_{j,2}^{*}$	$P(\{U_j \geq U_{-j}\} \cap \{\mathrm{EU}_{-j} \geq \mathrm{EU}_j\} \cap \{g_i(x_j, \epsilon_j, U_{-j}) \leq 0\})$	A2

TABLE A1 (*Continued*)

Notation	Definition	Section
$P^*_{j,3}$	$P(\{U_{-j} \geq U_j\} \cap \{\mathrm{EU}_j \geq \mathrm{EU}_{-j}\} \cap \{g_i(x_{-j}, \epsilon_{-j}, U_j) \leq 0\})$	A2
$P_{4new}, P^{\mathcal{S}}_{5new}, P^{\mathcal{S}}_{5new,1}, P^{\mathcal{S}}_{5new,2}$	Analog of P_4, $P^{\mathcal{S}}_5$, $P^{\mathcal{S}}_{5,1}$, $P^{\mathcal{S}}_{5,2}$ with observables impacting search but not utility	A5
$P_{1,sim}$	$P(U_1 \geq U_2)$	A7
$P_{2,sim}$	$P(\{U_1 > U_2\} \cap \{\mathrm{EU}_2 > \mathrm{EU}_1\} \cap \{g_{sim}(\mathrm{EU}_1 - \mathrm{EU}_2) < 0\})$	A7
$P_{4,\ out}, P^{\mathcal{S}}_{5,out}, P^{\mathcal{S}}_{5,1,out}, P^{\mathcal{S}}_{5,2,out}$	Analog of P_4, $P^{\mathcal{S}}_5$, $P^{\mathcal{S}}_{5,1}$, $P^{\mathcal{S}}_{5,2}$ with outside option available without search	A9
$P_{6,out}$	$P\ (\{U_1 \geq U_k\ \forall\ k \in \mathcal{J} \cup \{0\}\} \cap \{g(x_1, \epsilon_1, U_0) \leq 0\})$	A9

A1. *Proof of Lemma 2 When $J \geq 2$*

In this section, we prove lemma 2 for the more general case in which $J \geq 2$.

Let $\mathcal{J}_{-1} \equiv \{2, \ldots, J\}$, $\tilde{u}_j \equiv \alpha x_j + \beta z_j$ for all j, and $\tilde{\mathbf{u}} \equiv (\tilde{u}_1, \ldots, \tilde{u}_J)$. Similarly, let $\tilde{\mathrm{eu}}_j = \beta\gamma_0 + (\alpha + \beta\gamma_1)x_j$ and $\tilde{\mathbf{eu}} = (\tilde{\mathrm{eu}}_1, \ldots, \tilde{\mathrm{eu}}_J)$. Then, by (2) we can write for all $(\mathbf{x}, \mathbf{z})$ with $\tilde{z}_1 \geq \tilde{z}_j$ for all j:

$$\begin{aligned} s_1 &= P(U_1 \geq U_k\ \forall\ k) - \sum_{\mathcal{S} \subset \mathcal{J}_{-1}, \mathcal{S} \neq \varnothing} P(\{U_1 \geq U_k\ \forall\ k\} \cap \{\mathrm{EU}_j \geq \mathrm{EU}_1 \text{ for at least one } j \in \mathcal{S}\} \\ &\qquad \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \text{ for all } j \in \mathcal{S}\} \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \mathcal{J}_{-1} \backslash \mathcal{S}\}) \\ &\equiv P_4(\tilde{\mathbf{u}}) - \sum_{\mathcal{S} \subset \mathcal{J}_{-1}, \mathcal{S} \neq \varnothing} P^{\mathcal{S}}_5(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, x_1). \end{aligned} \tag{A1}$$

Further, for every $\mathcal{S} \subset \mathcal{J}_{-1}$, $\mathcal{S} \neq \varnothing$, we have

$$\begin{aligned} P^{\mathcal{S}}_5 &= P(\{U_1 \geq U_k\ \forall\ k\} \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \text{ for all } j \in \mathcal{S}\} \\ &\qquad \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \mathcal{J}_{-1} \backslash \mathcal{S}\}) \\ &\quad - P(\{U_1 \geq U_k\ \forall\ k\} \cap \{\mathrm{EU}_1 \geq \mathrm{EU}_j \text{ for all } j \in \mathcal{S}\} \\ &\qquad \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \text{ for all } j \in \mathcal{S}\} \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \mathcal{J}_{-1} \backslash \mathcal{S}\}) \\ &= P(\{U_1 \geq U_k\ \forall\ k\} \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \text{ for all } j \in \mathcal{S}\} \\ &\qquad \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \mathcal{J}_{-1} \backslash \mathcal{S}\}) \\ &\quad - P(\{\mathrm{EU}_1 \geq \mathrm{EU}_j \text{ for all } j \in \mathcal{S}\} \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \text{ for all } j \in \mathcal{S}\} \\ &\qquad \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \mathcal{J}_{-1} \backslash \mathcal{S}\}) \\ &\equiv P^{\mathcal{S}}_{5,1}(\tilde{\mathbf{u}}, x_1) - P^{\mathcal{S}}_{5,2}(\tilde{\mathbf{u}}_{-1}, \tilde{\mathbf{eu}}, x_1), \end{aligned}$$

where $\tilde{\mathbf{u}}_{-1} \equiv (\tilde{u}_2, \ldots, \tilde{u}_J)$. The first equality follows from basic set algebra, while the second follows from the fact that for all $j \in \mathcal{S}$ and all $k \in \mathcal{J}_{-1} \backslash \mathcal{S}$, (i) $\mathrm{EU}_1 \geq \mathrm{EU}_j$ implies that $U_1 \geq U_j$ since $\tilde{z}_1 \geq \tilde{z}_j$ for all $j \in \mathcal{J}_{-1}$ and (ii) $g(x_1, \epsilon_1, U_k) \geq 0 \geq g(x_1, \epsilon_1, U_j)$ implies that $U_k \leq U_j$, which (together with the implication in part i) implies that $U_1 \geq U_k$. Thus, the event $U_1 \geq U_k\ \forall\ k \in \mathcal{J}_{-1}$ is implied by the other events inside the probability and can be dropped. Now, note that $P^{\mathcal{S}}_{5,2}$ does not depend on z_1 and that $P^{\mathcal{S}}_{5,1}$—as well as P_4—depends only on x_j and z_j via $\tilde{u}_j$. Thus, the result follows from the chain rule. As in lemma 2, identification of the distribution of ϵ is obtained by considering choice sets in which $\tilde{z}_k$ is the same for all k and varying $\mathbf{x}$. QED

A2. *Proof of Lemma 3*

Let $P^*_{j,2}$ be the probability of failing to search—and thus choose—good j even if it is utility maximizing, and let $P^*_{j,3}$ be the probability of choosing j even when it is not utility maximizing (i.e., failing to search another, higher-utility good). Then, $s_j = P(U_j \geq U_k \ \forall\ k) - P^*_{j,2}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) + P^*_{j,3}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x})$, where $\tilde{u}_j = \alpha x_j + \beta z_j$, $\tilde{\mathbf{u}} \equiv (\tilde{u}_1, \dots, \tilde{u}_J)$, $\tilde{\mathrm{eu}}_j = \beta\gamma_0 + (\alpha + \beta\gamma_1)x_j$, and $\tilde{\mathbf{eu}} \equiv (\tilde{\mathrm{eu}}_1, \dots, \tilde{\mathrm{eu}}_J)$. Note that $P^*_{j,2}$ and $P^*_{j,3}$ depend on $\mathbf{x}$ via three channels: the deterministic part of the utilities $(\tilde{\mathbf{u}})$, the deterministic part of the expected utilities $(\tilde{\mathbf{eu}})$, and directly via the g_i function—that is, the decision of whether to search the various products. Differentiating, we obtain

$$\frac{\partial s_j}{\partial z_j} = \beta\left[\frac{\partial P(U_j \geq U_k \ \forall\ k)}{\partial \tilde{u}_j} - \frac{\partial P^*_{j,2}}{\partial \tilde{u}_j}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) + \frac{\partial P^*_{j,3}}{\partial \tilde{u}_j}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x})\right],$$

$$\begin{aligned}\frac{\partial s_j}{\partial x_j} = {} & \alpha\left[\frac{\partial P(U_j \geq U_k \ \forall\ k)}{\partial \tilde{u}_j} - \frac{\partial P^*_{j,2}}{\partial \tilde{u}_j}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) + \frac{\partial P^*_{j,3}}{\partial \tilde{u}_j}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x})\right] \\ & - \frac{\partial P^*_{j,2}}{\partial x_j}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) + \frac{\partial P^*_{j,3}}{\partial x_j}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) \\ & -(\alpha + \beta\gamma_1)\frac{\partial P^*_{j,2}}{\partial \tilde{\mathrm{eu}}_j}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) + (\alpha + \beta\gamma_1)\frac{\partial P^*_{j,3}}{\partial \tilde{\mathrm{eu}}_j}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}).\end{aligned}$$

Note that $(\partial P(U_j \geq U_{-j})/\partial \tilde{u}_j) - (\partial P^*_{j,2}/\partial \tilde{u}_j)(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) + (\partial P^*_{j,3}/\partial \tilde{u}_j)(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) = \partial s_j/\partial \tilde{u}_j \geq 0$.[34] Further, $(\partial P^*_{j,2}/\partial x_j)(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) \leq 0$ and $(\partial P^*_{j,3}/\partial x_j)(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) \geq 0$ due to our assumptions about the function g. Finally, $(\partial P^*_{j,2}/\partial \tilde{\mathrm{eu}}_j)(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) \leq 0$ and $(\partial P^*_{j,3}/\partial \tilde{\mathrm{eu}}_j)(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, \mathbf{x}) \geq 0$ by assumption 2(i). Therefore, normalizing $\alpha = 1$, we obtain $|(\partial s_j/\partial z_j)/(\partial s_j/\partial x_j)| \leq \beta$ under the assumption that $\alpha + \beta\gamma_1 \geq 0$. One can show that $|(\partial s_j/\partial z_k)/(\partial s_j/\partial x_k)| \leq \beta$ for $k \neq j$ using an analogous argument.

A3. *Identifying Good 1 When z_j Is Vector Valued in the Linear Homogeneous Case*

For simplicity, the results in the main text are for the case where z_j is scalar valued for all goods j. This implies that one can label good 1 as the good with the highest value of $\tilde{z}$ without loss of generality. As we have noted, if there are multiple z attributes per good, then our results apply if the data contain one choice set where one

[34] Increasing $\tilde{u}_j$ can only switch consumers from not choosing good j to choosing j but never the reverse. To see this, note first that, conditional on searching any given set of goods, increasing $\tilde{u}_j$ increases the probability that good j is chosen. Second, changing $\tilde{u}_j$ does not change the probability that good j is searched, which depends on $g_i(x_j, \epsilon_j, U_{-j})$ for each alternative searched good. Third, changing $\tilde{u}_j$ never makes other goods more likely to be searched. Specifically, an alternative good k is searched if and only if $g_i(x_k, \epsilon_k, U_{k'}) \geq 0$ for all goods k' currently searched. This quantity is unchanged for $k' \neq j$ and weakly decreasing for $k' = j$, so no good can become more likely to be searched. Therefore, $\partial s_j/\partial \tilde{u}_j \geq 0$.

good is preferable to all other goods on each of the $\tilde{z}$ attributes. This is not without loss.

The basic idea of our approach is that one can define a weighted average of the relevant z's where the weights are recoverable from first derivatives. Then we proceed as usual, defining good 1 as the good with the highest value of this weighted average. More formally, let z_{kj} be the kth hidden attribute of good j and let β_k be the associated preference parameter. As above, let $\tilde{u}_j = \alpha x_j + \beta z_j$. By assumption 2, we can write $s_j = f_{s_j}(\tilde{u}_1, \dots, \tilde{u}_J, x_1, \dots, x_J)$ for all j and thus $\partial s_j/\partial z_{kj} = (\partial f_{s_j}/\partial \tilde{u}_j)\beta_k$, implying that $(\partial s_j/\partial z_{kj})/(\partial s_j/\partial z_{k'j}) = \beta_k/\beta_{k'}$ for all k, k'. This means that we can compare the hidden component of utility across goods. Specifically, letting $\beta_1 > 0$ without loss, we have that, for any pair of goods j and j', $\Sigma_k \beta_k \tilde{z}_{kj} \geq \Sigma_k \beta_k \tilde{z}_{kj'}$ if and only if $\tilde{z}_{1j} - \tilde{z}_{1j'} + \Sigma_{k>1}(\beta_k/\beta_1)(\tilde{z}_{kj} - \tilde{z}_{kj'}) \geq 0$. Since the left-hand side of the last inequality is identified, we can rank goods based on the component of utility that is not known before search. Lemma 2 then applies by defining good 1 as the good with the highest value of $\Sigma_k \beta_k \tilde{z}_{kj}$. Note that such a good always exists in any choice set (excluding ties) since $\Sigma_k \beta_k \tilde{z}_{kj}$ is scalar valued.

A4. Endogenous Attributes

Here we show how to extend our results to the case where some product attributes are endogenous (sec. III.A). Consistent with standard results on nonparametric identification of demand with endogeneity (Berry and Haile 2014), we assume that the data contain information about choice probabilities across many markets. This is in contrast to the case without endogeneity, where just six markets are in principle sufficient. We start by considering the case where price is visible before search. Letting $\delta = (\delta_1, \dots, \delta_J)$, where δ_j is defined in assumption 3(i), and $\tilde{z}_j = z_j - (\gamma_0 + \gamma_1 x_j + \gamma_{1,p} p_j)$, we write the choice probability of good j as

$$s_j = \sigma_j(\delta, \mathbf{z}) \tag{A2}$$

for some function σ_j. Repeating this for all j and stacking the equations, we obtain a demand system of the form

$$\mathbf{s} = \sigma(\delta, \mathbf{z}), \tag{A3}$$

where $\mathbf{s} = (s_1, \dots, s_J)$. We also define the choice probability of the outside option as $s_0 \equiv 1 - \Sigma_{j=1}^J s_j$, with associated function $\sigma_0(\delta, \mathbf{z})$. We establish nonparametric identification of this demand system by invoking results from Berry and Haile (2014).[35] Specifically, the results in Berry and Haile (2014) yield identification of $(\xi_j)_{j=1}^J$ for every market in the population. This means that all the arguments of σ are known, which immediately implies (nonparametric) identification of σ itself. Once σ is identified, one may apply our results in section II.B to identify the preference parameters α, β, and λ. Note that, while knowledge of σ is sufficient for several counterfactuals of interest (e.g., computing equilibrium prices after a potential merger or tax), the preference parameters are required to predict how choices and welfare would change if consumers were given full information, among other things. In this sense, our approach complements the identification results in Berry and Haile (2014) within the class of search models we consider.

[35] See also Berry, Gandhi, and Haile (2013).

To prove identification of σ, we first note that model (A2) satisfies the index restriction in Berry and Haile's (2014) assumption 1. Second, we assume that we have excluded instruments $\mathbf{w}$ that, together with the exogenous attributes, satisfy the mean-independence restriction

$$E(\xi_j|\mathbf{x}, \mathbf{z}, \mathbf{w}) = 0 \quad \text{for all } j \tag{A4}$$

almost surely (assumption 3 in Berry and Haile 2014) and assume that the instruments shift the endogenous variables (choice probabilities and endogenous prices $\mathbf{p}$) to a sufficient degree (as in assumption 4 in Berry and Haile 2014). The endogeneity problem is exactly as in the full-information case; thus, all of the instruments that are commonly used (e.g., cost shifters, exogenous attributes of competing products, Hausman instruments) can be invoked in our context. Finally, we verify that the demand system satisfies the "connected substitutes" restriction defined in Berry and Haile's (2014) assumption 2. To this end, we prove the following result.

Lemma A1. Let utility be given by (8) with ϵ_i supported on $\mathbb{R}^J$, and let assumptions 2(i), 2(iii), 2(iv), and 3(i) hold. Then, for all j, $k = 1, \ldots, J$ with $j \neq k$, σ_j is (i) strictly increasing in δ_j and (ii) strictly decreasing in δ_k.

Proof. Fix (δ_j, z_j) for all j. To prove lemma A1(i), we show that an increase in δ_j can only induce a consumer to switch from not choosing j to choosing j, never vice versa, and that a positive mass of consumers will switch to choosing j. To see this, consider the case where consumer i initially searches j, which happens if and only if $g_i(\delta_j, \epsilon_{ij}, U_{ik}) \geq 0$ for all k such that $\mathrm{EU}_{ik} \geq \mathrm{EU}_{ij}$. Let $\Delta \geq 0$ be the change in δ_j. Since g_i is increasing in its first argument, we have $g_i(\delta_j + \Delta, \epsilon_{ij}, U_{ik}) \geq 0$ for all k such that $\mathrm{EU}_{ik} \geq \mathrm{EU}_{ij} + \Delta$ and thus i will still search j. Moreover, since g_i is decreasing in its last argument, if $g_i(\delta_k, \epsilon_{ik}, U_{ij}) \leq 0$ for some k such that $\mathrm{EU}_{ik} \leq \mathrm{EU}_{ij}$ (i.e., if k is initially not searched), then $g_i(\delta_k, \epsilon_{ik}, U_{ij} + \Delta) \leq 0$ (i.e., k is also not searched after the change in δ_j), which means that the set of goods searched by i never becomes larger. Next, note that if $U_{ij} \geq U_{ik}$ for all k in the set of searched goods $\mathcal{G}_i$, then $U_{ij} + \Delta \geq U_{ik}$ for all $k \in \mathcal{G}_i$. Further, since ϵ_i is supported on all of $\mathbb{R}^J$, there is a positive mass of consumers for which $U_{ik} \geq U_{ij}$ for some $k \in \mathcal{G}_i$, but $U_{ij} + \Delta \geq U_{ik}$ for all $k \in \mathcal{G}_i$. An analogous argument proves lemma A1(ii). QED

Lemma A1 implies that the goods are connected substitutes in δ (see definition 1 in Berry and Haile 2014), which in turn allows us to prove identification of σ by invoking theorem 1 in Berry and Haile (2014).[36] Specifically, we can invert the demand system σ for the indexes δ and write

$$\beta\gamma_0 + (\alpha + \beta\gamma_1)x_j + (\lambda + \beta\gamma_{1,p})p_j + \xi_j = \sigma_j^{-1}(\mathbf{s}, \mathbf{z}) \tag{A5}$$

for all j. Equations (A4) and (A5) naturally lead to a nonparametric instrumental variable approach to pin down σ_j^{-1} (and thus σ_j).[37]

[36] Note that the proof of theorem 1 in Berry and Haile (2014) uses only the fact that goods are connected substitutes in δ, not in $-\mathbf{p}$.

[37] Compiani (2022) proposes to approximate σ_j^{-1} using Bernstein polynomials. We use a similar approach in sec. V to estimate the demand function for the case without endogeneity.

Nothing in the argument above hinges on the fact that prices are visible to consumers before search. In particular, when consumers need to search to reveal prices, $z_j = p_j$ and equation (A3) becomes

$$\mathbf{s} = \sigma(\underline{\delta}, \mathbf{p}),$$

where $\underline{\delta}_j$ is defined in assumption 3(ii) and $\mathbf{p}$ denotes the vector of prices. Since the proof of lemma A1 continues to apply if assumption 3(i) is replaced with assumption 3(ii), one can follow the argument above to identify the function σ when prices are revealed only via search.

A5. *Identification When Observables Impact Search but Not Utility*

Here we state and prove the results described in section III.B. We make the following assumptions:

Assumption A1.

i) If consumer i searches j, then i also searches all j' such that $m(\mathrm{EU}_{ij'}, r_{j'}) \geq m(\mathrm{EU}_{ij}, r_j)$, where m is strictly increasing in both arguments.
ii) There is at least one good $j \neq 1$ such that $r_j > r_1$.
iii) The support of $(\mathbf{x}, \mathbf{z})|\mathbf{r}$ has positive Lebesgue measure for all $\mathbf{r} \equiv (r_1, \ldots, r_J)$.
iv) The search model admits a discrete choice representation that also satisfies the IIA property.

Assumption A1(iii) means that, for identification purposes, we consider variation in product characteristics holding fixed product search position. In practice, search position is likely to vary as a function of observables (e.g., products are sorted in order of price). However, because of the discrete nature of search position, we are likely to see variation conditional on search position and this is the variation we will use to identify our model. Assumption A1(iv) requires that consumers' search behavior can be represented as a standard discrete choice model satisfying IIA. As shown in Armstrong (2017), the Weitzman (1979) sequential search model (see example 1) can be represented as a discrete choice model where consumers maximize product-specific indexes defined as the minimum between the utility and the reservation value for each product. Then, assumption A1(iv) is satisfied by letting the ϵ_{ij} be Gumbel distributed.

If the order in which consumers search depends not only on the expected utilities but also on the variable r, lemma 1 will no longer hold as stated: the good with the highest value of $\tilde{z}_j$ can be searched, and another good j' may have higher utility (and thus higher expected utility), but good j' may not be searched because it has lower search position. However, an extension of lemma 1 will still hold in this case, which then allows us to prove identification of preferences.

Lemma A2. Let assumptions 1, 2(ii)–2(iv), and A1 hold. Then, if consumer i searches good 1 (i.e., the good with the highest value of $\tilde{z}$), then i chooses the good that maximizes utility among all goods with $r_j \geq r_1$.

Proof. Suppose there was a good j with $r_j \geq r_1$ and $U_{ij} > U_{i1}$ that consumer i does not search. We can follow the proof of lemma 1 to show that $\mathrm{EU}_{ij} > \mathrm{EU}_{i1}$. By assumption A1(i), this implies that good j is searched, which is a contradiction. QED

In words, if higher search position only makes a good more likely to be searched, then goods with higher utility and higher search position will always be searched if good 1 is searched. Given this lemma, we can apply a modification of the identification argument in lemma 2 after conditioning on the subset of goods with higher search position than good 1. As in lemma 2, we impose a location normalization ($\epsilon_{i\tilde{j}} = 0$ for some $\tilde{j}$ and all i) and a scale normalization ($\alpha = 1$).

LEMMA A3. Let the assumptions of lemma A2 hold. Let $s_{1|\mathcal{R}}$ denote the choice probability for good 1 conditional on consumers choosing in $\mathcal{R} = \{j : r_j \geq r_1\}$ and assume that $(\partial^2 s_{1|\mathcal{R}}/\partial z_1 \partial x_{j^*})(\mathbf{x}^*, \mathbf{z}^*, \mathbf{r}^*) \neq 0$ for some $(\mathbf{x}^*, \mathbf{z}^*, \mathbf{r}^*)$ and $j^* \neq 1$ and that $s_{1|\mathcal{R}}$ is twice differentiable. Then, β is identified. Additionally, the distribution of $\boldsymbol{\epsilon}$ is nonparametrically identified if the support of $(\epsilon_k - \epsilon_{\tilde{j}})_{k\neq\tilde{j}}$ is a subset of $\{(\alpha + \beta\gamma_1)(x_{\tilde{j}} - x_k)_{k\neq\tilde{j}} : \gamma_1(x_{\tilde{j}} - x_k)_{k\neq\tilde{j}} = (z_{\tilde{j}} - z_k)_{k\neq\tilde{j}}$ for some $(\mathbf{x}, \mathbf{z})$ in its support$\}$, and the supports of $\tilde{\mathbf{z}}$ and $\mathbf{r}$ contain a point such that $\tilde{z}_j = \tilde{z}_k$ and $r_j = r_k$ for all j, k, respectively.

Proof. Under assumption A1(iv), the choice probability for good 1 conditional on consumers choosing in $\mathcal{R}$, $s_{1|\mathcal{R}}$, is equal to the choice probability for good 1 if consumers faced only $\mathcal{R}$ as their choice set. Further, by lemma A2, the only mistake a consumer can make when faced with choice set $\mathcal{R}$ is to fail to search good 1 when it is in fact the good with the highest utility in $\mathcal{R}$. Thus, letting $\mathcal{R}_{-1} = \mathcal{R}\setminus\{1\}$, we can write for all $(\mathbf{x}, \mathbf{z})$ with $\tilde{z}_1 \geq \tilde{z}_j$ for all j:

$$\begin{aligned}
s_{1|\mathcal{R}} &= P(U_1 \geq U_k \ \forall\ k \in \mathcal{R}) \\
&\quad - \sum_{\mathcal{S}\subset\mathcal{R}_{-1},\mathcal{S}\neq\varnothing} P(\{U_1 \geq U_k \ \forall\ k \in \mathcal{R}\} \cap \{m(\mathrm{EU}_j, r_j) \geq m(\mathrm{EU}_1, r_1) \text{ for some } j \in \mathcal{S}\} \\
&\quad \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \text{ for all } j \in \mathcal{S}\} \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \mathcal{R}_{-1}\setminus\mathcal{S}\}) \\
&\equiv P_{4new}(\tilde{\mathbf{u}}) - \sum_{\mathcal{S}\subset\mathcal{R}_{-1},\mathcal{S}\neq\varnothing} P^{\mathcal{S}}_{5new}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, x_1, \mathbf{r}).
\end{aligned} \tag{A6}$$

Further, for every $\mathcal{S} \subset \mathcal{R}_{-1}$, $\mathcal{S} \neq \varnothing$, we have

$$\begin{aligned}
P^{\mathcal{S}}_{5new} &= P(\{U_1 \geq U_k \ \forall\ k \in \mathcal{R}\} \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \text{ for all } j \in \mathcal{S}\} \\
&\qquad \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \mathcal{R}_{-1}\setminus\mathcal{S}\}) \\
&\quad - P(\{U_1 \geq U_k \ \forall\ k \in \mathcal{R}\} \cap \{m(\mathrm{EU}_1, r_1) \geq m(\mathrm{EU}_j, r_j) \text{ for all } j \in \mathcal{S}\} \\
&\qquad \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \text{ for all } j \in \mathcal{S}\} \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \mathcal{R}_{-1}\setminus\mathcal{S}\}) \\
&= P(\{U_1 \geq U_k \ \forall\ k \in \mathcal{R}\} \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \text{ for all } j \in \mathcal{S}\} \\
&\qquad \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \mathcal{R}_{-1}\setminus\mathcal{S}\}) \\
&\quad - P(\{m(\mathrm{EU}_1, r_1) \geq m(\mathrm{EU}_j, r_j) \ \ \forall\ j \in \mathcal{S}\} \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \ \forall\ j \in \mathcal{S}\} \\
&\qquad \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \ \forall\ j \in \mathcal{R}_{-1}\setminus\mathcal{S}\}) \\
&\equiv P^{\mathcal{S}}_{5new,1}(\tilde{\mathbf{u}}, x_1, \mathbf{r}) - P^{\mathcal{S}}_{5new,2}(\tilde{\mathbf{u}}_{-1}, \tilde{\mathbf{eu}}, x_1, \mathbf{r}),
\end{aligned}$$

where $\tilde{\mathbf{u}}_{-1} \equiv (\tilde{u}_2, \ldots, \tilde{u}_J)$. The first equality follows from basic set algebra, while the second follows from the fact that the event $\{U_1 \geq U_j \ \forall\ j \in \mathcal{R}\}$ is implied by the other events inside the second probability, since (i) if $j \in \mathcal{S}$, then $m(\mathrm{EU}_1, r_1) \geq m(\mathrm{EU}_j, r_j)$ implies that $\mathrm{EU}_1 \geq \mathrm{EU}_j$, which in turn implies that $U_1 \geq U_j$, and (ii) if $j \notin \mathcal{S}$, then $g(x_1, \epsilon_1, U_j) \geq 0 \geq g(x_1, \epsilon_1, U_k)$ for all $k \in \mathcal{S}$ implies that $U_j \leq U_k$. Note that $P^{\mathcal{S}}_{5new,2}$ does not depend on z_1 and $P^{\mathcal{S}}_{5new,1}(\tilde{\mathbf{u}}, x_1, \mathbf{r})$ depends on x_j and z_j only via $\tilde{u}_j$ for $j \neq 1$, so that $(\partial^2 s_{1|\mathcal{R}}/\partial z_1 \partial z_{j^*})(\mathbf{x}^*, \mathbf{z}^*, \mathbf{r}^*)/(\partial^2 s_{1|\mathcal{R}}/\partial z_1 \partial x_{j^*})(\mathbf{x}^*, \mathbf{z}^*, \mathbf{r}^*) = \beta/\alpha$. Note that $s_{1|\mathcal{R}}$ can be estimated by taking $\mathcal{R}$ as the choice set faced by consumers and dropping those consumers who choose products outside $\mathcal{R}$. Thus, β/α is identified, as well as β given the normalization $\alpha = 1$. Finally, we identify the distribution of ϵ by looking at choice sets in which $\tilde{z}_j = \tilde{z}$ and $r_j = r$ for all j. Consumers facing such choice sets always choose the utility-maximizing good, since the first good they search—the one that maximizes $m(\mathrm{EU}_j, r_j)$—also maximizes utility. Thus, by varying $\mathbf{x}$, we can identify the distribution of ϵ just like in lemma 2. Again, because this argument conditions on the slice of the data where $\tilde{z}_j = \tilde{z}$ and $r_j = r$ for all j, it is likely to lead to slower than parametric convergence rates (Khan and Tamer 2010). QED

A6. Unobservables Revealed by Search

Here we show that the ratio of second derivatives in (3) robustly identifies β/α in the model where ϵ_{ij} is revealed to consumer i only upon searching good j (sec. III.C). Order goods in increasing order of x. Then, if $\alpha + \beta\gamma_1 \geq 0$, consumers search in descending order of x. Thus, for $j = 1, \ldots, J$,

$$\begin{aligned} s_j &= \sum_{k=1}^{j} P(\{U_j \geq U_{j'} \ \forall\ j' \in \{k, \ldots, J\}\} \cap \{\text{search exactly } k, \ldots, J\}) \\ &= \sum_{k=1}^{j} P(\{U_j \geq U_{j'} \ \forall\ j' \in \{k, \ldots, J\}\} \cap \{g(x_h, U_{h'}) \geq 0 \ \forall\ h = k, \ldots, J-1;\ h' \in \{h+1, \ldots, J\}\} \cap \\ &\quad \{g(x_h, U_j) \leq 0 \ \forall\ h = 1, \ldots, k-1\}) \\ &\equiv \sum_{k=1}^{j} P_j^{(k)}(\tilde{\mathbf{u}}, \mathbf{x}_{-J}), \end{aligned}$$

where $\tilde{u}_j = \alpha x_j + \beta z_j$ and $\tilde{\mathbf{u}} = (\tilde{u}_1, \ldots, \tilde{u}_J)$, as above, and $\mathbf{x}_{-J} = (x_1, \ldots, x_{J-1})$. Thus,

$$\begin{aligned} \frac{\partial^2 s_j}{\partial z_j \partial z_J} &= \sum_{k=1}^{j} \frac{\partial^2 P_j^{(k)}}{\partial \tilde{u}_j \partial \tilde{u}_J} \beta^2, \\ \frac{\partial^2 s_j}{\partial z_j \partial x_J} &= \sum_{k=1}^{j} \frac{\partial^2 P_j^{(k)}}{\partial \tilde{u}_j \partial \tilde{u}_J} \alpha\beta. \end{aligned}$$

Thus, the ratio of the latter two derivatives identifies β/α. Note that the ratio of $\partial s_j/\partial z_J$ to $\partial s_j/\partial x_J$ for any j would also work, thus providing a test for the null hypothesis that ϵ is revealed only via search. Alternatively, for $j < J$,

$$\begin{aligned} \frac{\partial s_j}{\partial z_j} &= \sum_{k=1}^{j} \frac{\partial P_j^{(k)}}{\partial \tilde{u}_j} \beta, \\ \frac{\partial s_j}{\partial x_j} &= \sum_{k=1}^{j} \left(\frac{\partial P_j^{(k)}}{\partial \tilde{u}_j} + \frac{1}{\alpha} \frac{\partial P_j^{(k)}}{\partial x_j}\right) \alpha. \end{aligned} \tag{A7}$$

Since $(1/\alpha)(\partial P_j^{(k)}/\partial x_j) \geq 0$, (A7) implies that the ratio of first derivatives suffers from attenuation bias—that is,

$$\frac{\partial s_j/\partial z_j}{\partial s_j/\partial x_j} \leq \frac{\beta}{\alpha}.$$

A7. Simultaneous Search à la Chade and Smith (2006)

While our main model allows for consumers to choose which goods to search in one simultaneous step, one form of simultaneous search that is not accommodated is that in which a consumer optimally chooses the number K of goods to uncover and then proceeds to simultaneously search the top K in terms of expected utility (see Chade and Smith 2006; for an application, see Honka, Hortaçsu, and Vitorino 2017). Our framework from section II does not subsume this model since in this case the decision of whether to search good j depends not only on the expected utility of good j but on the expected utility of all other goods as well, thus violating assumption 2(ii).

In this appendix, we show via an alternative argument that the usual second-derivative ratio from equation (3) still identifies β/α in the two-good K-rank model. This complements the simulation results from section IV of the supplementary appendix, showing that our method succeeds in a model where consumers search the top K goods (with K varying randomly across consumers).

Consider the simultaneous search model in Honka, Hortaçsu, and Vitorino (2017) with $J = 2$ goods. In this model, a consumer looks at the expected utilities and decides whether to only search the good with the highest expected utility or search both goods, which entails a cost c. Consumers form expectations over the distribution of $(\tilde{z}_1, \tilde{z}_2)$, assumed to be independent of everything else. As usual, we denote by 1 the good with the highest value of $\tilde{z}$.

Note that consumer i searches 2 but not 1 if and only if $\mathrm{EU}_{i2} > \mathrm{EU}_{i1}$ and

$$E_{\tilde{z}_1,\tilde{z}_2}[\max\{\mathrm{EU}_{i1} + \beta\tilde{z}_1, \mathrm{EU}_{i2} + \beta\tilde{z}_2\}] - c < E_{\tilde{z}_2}[\mathrm{EU}_{i2} + \beta\tilde{z}_2];$$

that is,

$$E_{\tilde{z}_1,\tilde{z}_2}[\max\{\mathrm{EU}_{i1} - \mathrm{EU}_{i2} + \beta(\tilde{z}_1 - \tilde{z}_2), 0\}] - c < 0$$

or $g_{sim}(\mathrm{EU}_{i1} - \mathrm{EU}_{i2}) < 0$ for an increasing function g_{sim}. Equation (2) can then be written as

$$\begin{aligned} s_1 &= P(U_1 > U_2) - P(\{U_1 > U_2\} \cap \{\mathrm{EU}_2 > \mathrm{EU}_1\} \cap \{g_{sim}(\mathrm{EU}_1 - \mathrm{EU}_2) < 0\}) \\ &= P_{1,sim} - P_{2,sim}. \end{aligned}$$

Letting $\tilde{u}_j = \alpha x_j + \beta z_j$ and $\tilde{\mathrm{eu}}_j = \beta\gamma_0 + (\alpha + \beta\gamma_1)x_j$, we also have

$$\frac{\partial^2 s_1}{\partial z_1 \partial z_2} = \beta^2 \left(\frac{\partial^2 P_{1,sim}}{\partial \tilde{u}_1 \partial \tilde{u}_2} - \frac{\partial^2 P_{2,sim}}{\partial \tilde{u}_1 \partial \tilde{u}_2} \right) \tag{A8}$$

and

$$\frac{\partial^2 s_1}{\partial z_1 \partial x_2} = \alpha\beta \left(\frac{\partial^2 P_{1,sim}}{\partial \tilde{u}_1 \partial \tilde{u}_2} - \frac{\partial^2 P_{2,sim}}{\partial \tilde{u}_1 \partial \tilde{u}_2} \right) - (\alpha + \beta\gamma_1)\beta \frac{\partial^2 P_{2,sim}}{\partial \tilde{u}_1 \partial \tilde{\mathrm{eu}}_2}. \tag{A9}$$

So, if $\partial^2 P_{2,sim}/\partial \tilde{u}_1 \partial \mathrm{e\tilde{u}}_2 = 0$, then the ratio of (A8) to (A9) identifies β/α. Note that the event in $P_{2,sim}$ is equivalent to the following set of inequalities: (i) $\epsilon_{i1} > \tilde{u}_2 - \tilde{u}_1 + \epsilon_{i2}$, (ii) $\epsilon_{i1} < \mathrm{e\tilde{u}}_2 - \mathrm{e\tilde{u}}_1 + \epsilon_{i2}$, (iii) $\epsilon_{i1} < g_{sim}^{-1}(0) + \mathrm{e\tilde{u}}_2 - \mathrm{e\tilde{u}}_1 + \epsilon_{i2}$. Then, letting $\tilde{\epsilon} = \epsilon_1 - \epsilon_2$, we have

$$P_{2,sim} = \int_{\tilde{u}_2 - \tilde{u}_1}^{\min(\mathrm{e\tilde{u}}_2 - \mathrm{e\tilde{u}}_1,\, g_{sim}^{-1}(0) + \mathrm{e\tilde{u}}_2 - \mathrm{e\tilde{u}}_1)} f_{\tilde{\epsilon}}(\tilde{\epsilon})\, d\tilde{\epsilon}. \tag{A10}$$

By Leibniz's rule, $\partial P_{2,sim}/\partial \tilde{u}_1 = f_{\tilde{\epsilon}}(\tilde{u}_2 - \tilde{u}_1)$ and thus $\partial^2 P_{2,sim}/\partial \tilde{u}_1 \partial \mathrm{e\tilde{u}}_2 = 0$.

Finally, we show that the ratio of first derivatives leads to attenuation bias if $\alpha + \beta\gamma_1 \geq 0$. This follows directly from

$$\begin{aligned}
\frac{\partial s_1}{\partial z_1} &= \beta \left(\frac{\partial P_{1,sim}}{\partial \tilde{u}_1} - \frac{\partial P_{2,sim}}{\partial \tilde{u}_1} \right), \\
\frac{\partial s_1}{\partial x_1} &= \alpha \left(\frac{\partial P_{1,sim}}{\partial \tilde{u}_1} - \frac{\partial P_{2,sim}}{\partial \tilde{u}_1} - \frac{\alpha + \beta\gamma_1}{\alpha} \frac{\partial P_{2,sim}}{\partial \mathrm{e\tilde{u}}_1} \right),
\end{aligned}$$

and the fact that $\partial P_{2,sim}/\partial \mathrm{e\tilde{u}}_1 < 0$.

A8. Discrete Attributes

Our main result applies to the case where one can take derivatives of choice probabilities with respect to the attribute of various goods, implying that the attributes must be continuously distributed. Here we show that our argument can be extended to the case where attributes are discrete provided derivatives are replaced with differences and at least one attribute is continuous. We begin with the case where z is discrete and there is at least one continuous x attribute (additional discrete x attributes are allowed).

Let 1 denote one of the goods with the highest value of $\tilde{z}$ in the choice set. When z is discrete, there can be multiple such goods, but the argument holds for any one of them. Take any data point $(\mathbf{x}, \mathbf{z})$ and let $D_{\delta_{z_1}, z_1} s_1(\mathbf{x}, \mathbf{z}) \equiv s_1(\mathbf{x}, \mathbf{z} + \delta_{z_1}) - s_1(\mathbf{x}, \mathbf{z})$, where $\mathbf{z} + \delta_{z_1}$ is equal to $\mathbf{z}$ except that z_1 is increased or decreased by δ_{z_1}. By the same argument as in the proof of lemma 2, $D_{\delta_{z_1}, z_1} s_1(\mathbf{x}, \mathbf{z})$ depends on x_j and z_j for $j \neq 1$ only via $\alpha x_j + \beta z_j$ (this relies on lemma 1, which holds regardless of whether z is discrete or continuous). Thus, as in the continuous case, the marginal rate of substitution β/α will be identified by looking at how $D_{\delta_{z_1}, z_1} s_1(\mathbf{x}, \mathbf{z})$ changes as z_j and x_j vary for $j \neq 1$. Specifically, take a change from z_j to $z_j + \delta_{z_j}$ and a change from x_j to $x_j + \delta_{x_j}$ that lead to the same change in $\tilde{u}_j \equiv \alpha x_j + \beta z_j$. It must be that $\beta \delta_{z_j} = \alpha \delta_{x_j}$, implying that $\beta/\alpha = \delta_{x_j}/\delta_{z_j}$. Thus, given a discrete change δ_{z_j}, finding the corresponding change δ_{x_j} identifies the marginal rate of substitution. To this end, we look at how $D_{\delta_{z_1}, z_1} s_1(\mathbf{x}, \mathbf{z})$ changes as x_j moves continuously. The change δ_{x_j} is a solution of the following equation in d_{x_j}:

$$D_{\delta_{z_1}, z_1} s_1(\mathbf{x} + d_{x_j}, \mathbf{z}) = D_{\delta_{z_1}, z_1} s_1(\mathbf{x}, \mathbf{z} + \delta_{z_j}), \tag{A11}$$

where $\mathbf{z} + \delta_{z_j}$ is equal to $\mathbf{z}$ except that z_j is changed to $z_j + \delta_{z_j}$ and similarly for $\mathbf{x} + d_{x_j}$. If $D_{\delta_{z_1}, z_1} s_1(\mathbf{x}, \mathbf{z})$ is monotonic in $\tilde{u}_j$, δ_{x_j} is the unique solution of (A11). More generally, we show in the following lemma that the system of equations given by (A11) for all $\mathbf{x}$, $\mathbf{z}$ has a unique solution under a regularity condition ruling out "pathological cases."

Definition A1. A function $f : (a, b) \to \mathbb{R}$ is nonperiodic if there is no $K \in (0, b - a)$ such that $f(\tilde{u} + K) = f(\tilde{u})$ for all $\tilde{u} \in (a, b - K)$.

Lemma A4. Let (a, b) be the support of $\alpha x_j + \beta z_j$, where a and/or b may be infinite. Assume that $D_{\delta_{z_1}, z_1} s_1$ is a nonperiodic function of $\alpha x_j + \beta z_j$ for some $j \neq 1$. Then $\beta/\alpha = \beta$ is point identified.

Proof. Since $D_{\delta_{z_1}, z_1} s_1$ depends on x_j and z_j only via $\tilde{u}_j \equiv \alpha x_j + \beta z_j$ for $j \neq 1$, fixing all other arguments, we can write $D_{\delta_{z_1}, z_1} s_1(\mathbf{x}, \mathbf{z}) = h(\tilde{u}_j)$. Consider a discrete increase δ_{z_j} in z_j, and note that an increase in x_j by $\delta_{x_j} \equiv (\beta/\alpha)\delta_{z_j}$ will deliver the same change in h. Suppose by contradiction that there is a $\delta'_{x_j} \neq (\beta/\alpha)\delta_{z_j}$ that delivers the same change in h for any baseline level of $\tilde{u}_j$. First, take the case where $\alpha\delta'_{x_j} > \beta\delta_{z_j}$. Then, we have $h(\tilde{u}_j + \alpha\delta'_{x_j}) = h(\tilde{u}_j + \beta\delta_{z_j})\ \forall\ \tilde{u}_j \in (a - \beta\delta_{z_j}, b - \alpha\delta'_{x_j})$, which holds if and only if $h(\tilde{u}_j + \alpha\delta'_{x_j} - \beta\delta_{z_j}) = h(\tilde{u}_j)\ \forall\ \tilde{u}_j \in (a, b - (\alpha\delta'_{x_j} - \beta\delta_{z_j}))$. This contradicts the assumption that h is nonperiodic with $K = \alpha\delta'_{x_j} - \beta\delta_{z_j}$. The case where $\alpha\delta'_{x_j} < \beta\delta_{z_j}$ is symmetric. Specifically, we have $h(\tilde{u}_j + \alpha\delta'_{x_j}) = h(\tilde{u}_j + \beta\delta_{z_j})$ $\forall\ \tilde{u}_j \in (a - \alpha\delta'_{x_j}, b - \beta\delta_{z_j})$, which holds if and only if $h(\tilde{u}_j) = h(\tilde{u}_j + \beta\delta_{z_j} - \alpha\delta'_{x_j})$ $\forall\ \tilde{u}_j \in (a, b - (\beta\delta_{z_j} - \alpha\delta'_{x_j}))$. This contradicts the assumption that h is nonperiodic with $K = \beta\delta_{z_j} - \alpha\delta'_{x_j}$. Thus, β/α is uniquely identified as $\delta_{x_j}/\delta_{z_j}$. QED

The identification argument above suggests the following estimation procedure:

1) Estimate the choice probability function for good 1, s_1. (Here we focus on the case where one is using the nonparametric method of sec. V.A, but alternative methods, such as flexible logit, could be used as well.)
2) For every choice set $(\mathbf{x}, \mathbf{z})$ in the data, evaluate $s_1(\mathbf{x}, \mathbf{z} + \delta_{z_1} + \delta_{z_j}) - s_1(\mathbf{x}, \mathbf{z} + \delta_{z_j})$ for given increments δ_{z_1} and δ_{z_j}. (Consistent with the notation above, we let $\mathbf{z} + \delta_{z_1} + \delta_{z_j}$ denote a vector equal to $\mathbf{z}$ except that z_1 and z_j are increased by δ_{z_1} and δ_{z_j}, respectively.)
3) Evaluate $s_1(\mathbf{x} + \delta_{x_j}, \mathbf{z} + \delta_{z_1}) - s_1(\mathbf{x} + \delta_{x_j}, \mathbf{z})$, and repeat this by varying δ_{x_j} over a grid of points. Find the value of δ_{x_j} that makes this difference as close as possible to the difference $s_1(\mathbf{x}, \mathbf{z} + \delta_{z_1} + \delta_{z_j}) - s_1(\mathbf{x}, \mathbf{z} + \delta_{z_j})$ from the previous step. Call this value $\bar{\delta}_{x_j}$.
4) Repeat steps 2 and 3 for every choice set $(\mathbf{x}, \mathbf{z})$ in the data, and take the ratio of a trimmed mean of the $\bar{\delta}_{x_j}$ values to δ_{z_j}. This is an estimate of β/α.

The simulations in section IV of the supplementary appendix suggest that this procedure delivers consistent estimates of β/α.

Note that the argument above continues to hold if we instead assume that z is continuous and x is discrete. Specifically, by lemma 1, we have that $\partial s_1/\partial z_1$ depends on x_j and z_j only via $\alpha x_j + \beta z_j$. Thus, assuming that $\partial s_1/\partial z_1$ is a nonperiodic function of $\alpha x_j + \beta z_j$, the argument in lemma A4 with the roles of x_j and z_j flipped shows identification of β/α when z is continuous and x is discrete.

Finally, regardless of whether x and z are continuous or discrete, conditioning on $\tilde{z}_j$ being the same across goods implies that consumers always maximize utility. Thus, we can continue to invoke standard full-information arguments to identify the distribution of ϵ.

A9. Outside Option Known for Free

Here we show that lemma 2 can be extended to accommodate an outside option whose utility, U_0, is known for free by consumers. Because lemma 1 still applies in this case, we can adapt the proof in appendix section A.1 as follows:

$$\begin{aligned}
s_1 &= P(U_1 \geq U_k \ \forall\ k \in \mathcal{J} \cup \{0\}) \\
&\quad - \sum_{\mathcal{S} \subset \mathcal{J}_{-1}, \mathcal{S} \neq \varnothing} P(\{U_1 \geq U_k \ \forall\ k \in \mathcal{J} \cup \{0\}\} \cap \{\mathrm{EU}_j \geq \mathrm{EU}_1 \text{ for at least one } j \in \mathcal{S}\} \\
&\quad \cap \{g(x_1, \epsilon_1, U_j) \leq 0 \text{ for all } j \in \mathcal{S}\} \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \{\mathcal{J}_{-1} \backslash \mathcal{S}\} \cup \{0\}\}) \\
&\quad - P(\{U_1 \geq U_k \ \forall\ k \in \mathcal{J} \cup \{0\}\} \cap \{g(x_1, \epsilon_1, U_0) \ \leq \ 0\}) \\
&\equiv P_{4,out}(\tilde{\mathbf{u}}) - \sum_{\mathcal{S} \subset \mathcal{J}_{-1}, \mathcal{S} \neq \varnothing} P^{\mathcal{S}}_{5,out}(\tilde{\mathbf{u}}, \tilde{\mathbf{eu}}, x_1) - P_{6,out}(\tilde{\mathbf{u}}, x_1).
\end{aligned}$$

The terms $P^{\mathcal{S}}_{5,out}$ correspond to the case where consumers fail to search good 1 because of a (relatively) high-utility inside product, whereas $P_{6,out}$ corresponds to the case where they fail to search good 1 due to the outside option. Next, for every $\mathcal{S} \subset \mathcal{J}_{-1}$, $\mathcal{S} \neq \varnothing$, we have

$$\begin{aligned}
P^{\mathcal{S}}_{5,out} &= P(\{U_1 \geq U_k \ \forall\ k \in \mathcal{J} \cup \{0\}\} \cap \{g(x_1, \epsilon_1, U_j) \ \leq \ 0 \text{ for all } j \in \mathcal{S}\} \\
&\qquad \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \{\mathcal{J}_{-1} \backslash \mathcal{S}\} \cup \{0\}\}) \\
&\quad - P(\{U_1 \geq U_k \ \forall\ k \in \mathcal{J} \cup \{0\}\} \cap \{\mathrm{EU}_1 \geq \mathrm{EU}_j \text{ for all } j \in \mathcal{S}\} \\
&\qquad \cap \{g(x_1, \epsilon_1, U_j) \ \leq \ 0 \text{ for all } j \in \mathcal{S}\} \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \{\mathcal{J}_{-1} \backslash \mathcal{S}\} \cup \{0\}\}) \\
&= P(\{U_1 \geq U_k \ \forall\ k \in \mathcal{J} \cup \{0\}\} \cap \{g(x_1, \epsilon_1, U_j) \ \leq \ 0 \text{ for all } j \in \mathcal{S}\} \\
&\qquad \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \{\mathcal{J}_{-1} \backslash \mathcal{S}\} \cup \{0\}\}) \\
&\quad - P(\{\mathrm{EU}_1 \geq \mathrm{EU}_j \text{ for all } j \in \mathcal{S}\} \cap \{g(x_1, \epsilon_1, U_j) \ \leq \ 0 \text{ for all } j \in \mathcal{S}\} \\
&\qquad \cap \{g(x_1, \epsilon_1, U_j) \geq 0 \text{ for all } j \in \{\mathcal{J}_{-1} \backslash \mathcal{S}\} \cup \{0\}\}) \\
&\equiv P^{\mathcal{S}}_{5,1,out}(\tilde{\mathbf{u}}, x_1) - P^{\mathcal{S}}_{5,2,out}(\tilde{\mathbf{u}}_{-1}, \tilde{\mathbf{eu}}, x_1),
\end{aligned}$$

where the derivation follows from the same argument as in appendix section A1. The fact that $P_{4,out}$, $P^{\mathcal{S}}_{5,1,out}$, and $P_{6,out}$ depend on x_j and z_j only via $\tilde{u}_j$ and that $P^{\mathcal{S}}_{5,2,out}$ does not depend on z_1 implies that we can identify β using the same ratio of second derivatives as in lemma 2.

To recover the distribution of ϵ (or, if the latter is assumed to be known, the coefficient α), we need to isolate consumers who choose the utility-maximizing good. As in the baseline model, we achieve this by conditioning on choice sets where $\tilde{z}_j = \tilde{z}$ for all j. Under the assumption that consumers always search at least one of the inside goods, our model implies that consumers always search the good with the highest expected utility, which is also the good with the highest realized utility when $\tilde{z}_j$ is the same for all j. Thus, full-information arguments can be invoked to trace out the distribution of ϵ as in the baseline case.

To shed some light on the bias arising when this assumption is not satisfied, we consider the case where the distribution of ϵ is assumed to be known and the researcher uses the argument in the above paragraph to identify α. Conditioning on $\tilde{z}_j = \tilde{z}$ for all $j \in \mathcal{J}$, we can write

$$\begin{aligned} s_j &= P(\{U_j \geq U_k \ \forall\ k \in \mathcal{J} \cup \{0\}\}) \\ &\quad - P(\{U_j \geq U_k \ \forall\ k \in \mathcal{J} \cup \{0\}\} \cap \{\text{no inside good is searched}\}) \\ &\equiv P_{7,out} - P_{8,out}. \end{aligned} \tag{A12}$$

These probabilities depend on the specific way in which consumers search, something that our model intentionally does not fully specify. Thus, it is hard to quantify or even sign the bias coming from erroneously maintaining the "one free search" assumption without additional structure placed on the model. Nonetheless, we can use equation (A12) to isolate different sources of bias and show that, in special cases, this can be used to sign the bias. First, note that our model yields

$$P_{8,out} = P(\{U_j \geq U_k \ \forall\ k \in \mathcal{J} \cup \{0\}\} \cap \{g(x_h, \epsilon_h, U_0) \ \leq \ 0 \ \forall\ h \in \mathcal{J}\}).$$

Further,

$$\frac{\partial s_j}{\partial x_j} = \left(\frac{\partial P_{7,out}}{\partial \tilde{u}_j} - \frac{\partial P_{8,out}}{\partial \tilde{u}_j}\right)\alpha - \frac{\partial P_{8,out}}{\partial x_j}, \tag{A13}$$

where $\tilde{u}_j \equiv \alpha x_j + \beta z_j$. Now, we want to quantify the bias in estimating α from incorrectly assuming that $P_{8,out} \equiv 0$ (in what follows, we focus on the bias arising from incorrectly specifying the model and we abstract away from sampling error, so any statements about bias should be interpreted as holding in large samples). If one (incorrectly) assumes that $P_{8,out} \equiv 0$, α is identified by relating $\partial s_j/\partial x_j$ to $\partial P_{7,out}/\partial \tilde{u}_j$. For example, in a logit model, $\partial s_j/\partial x_j = (\partial P_{7,out}/\partial \tilde{u}_j)\alpha$ with $\partial P_{7,out}/\partial \tilde{u}_j = s_j(1 - s_j)$, so that α is identified as $(\partial s_j/\partial x_j)/(\partial P_{7,out}/\partial \tilde{u}_j)$. When applying this approach to data where in reality consumers behave as in equation (A13), we have

$$\frac{\partial s_j}{\partial x_j} \Big/ \frac{\partial P_{7,out}}{\partial \tilde{u}_j} = \left(1 - \frac{\partial P_{8,out}}{\partial \tilde{u}_j} \Big/ \frac{\partial P_{7,out}}{\partial \tilde{u}_j}\right)\alpha - \frac{\partial P_{8,out}}{\partial x_j} \Big/ \frac{\partial P_{7,out}}{\partial \tilde{u}_j}. \tag{A14}$$

Since $\partial P_{8,out}/\partial \tilde{u}_j \geq 0$ and (under standard regularity conditions on the distribution of ϵ) $\partial P_{7,out}/\partial \tilde{u}_j > 0$, the coefficient on α in (A14) is ≤ 1. If this was the only channel through which bias arises, incorrectly ignoring that consumers may sometimes not search any of the inside goods would lead to downward bias in the estimate of α. However, the term $(\partial P_{8,out}/\partial x_j)/(\partial P_{7,out}/\partial \tilde{u}_j)$ in (A14) complicates things. Under reasonable search protocols, we will typically have $\partial P_{8,out}/\partial x_j \leq 0$ (i.e., making good j's expected utility more appealing decreases the probability that a consumer does not search any of the inside goods; we already made this assumption in lemma 3). Thus, the term $-((\partial P_{8,out}/\partial x_j)/(\partial P_{7,out}/\partial \tilde{u}_j))$ in (A14) is positive, which tends to inflate the estimate of α. Which of these two effects dominates ultimately depends on the specific search protocol. For example, if x_j does not affect the initial decision of whether to search any of the inside goods (as in the case of satisficing from example 3 in the main text), then $\partial P_{8,out}/\partial x_j = 0$ and the estimate of α will tend to be biased downward.

Data Availability

Data and replication codes can be found in Abaluck, Compiani, and Zhang (2025) in the Harvard Dataverse, https://doi.org/10.7910/DVN/JCXVD9. This contains a detailed README file indicating which codes need to be run for each table and figure in the paper.

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